Master's Thesis

Stochastic Dynamics of Bacterial Type-IV Pili

Stochastische Dynamik der bakteriellen Typ-IV-Pili

Department of Physics
Ludwig-Maximilians-Universität München

Kaan Bozgül

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Abstract

Type-IV pili (T4P) are extracellular appendages that can be used for many functions such as natural transformation, adhesion and twitching motility. This wide range of usage gives rise to a necessity to understand their dynamics and interactions for many use cases like antibiotic resistance, virulence, biofilm formation and bacterial adaptation to various environments. In this thesis, the ability of T4P to uptake molecules (such as DNA) from their environment, as well as their capacity to bind to surfaces, was simulated and quantified using the stochastic binding model of motor proteins. For the molecule uptake, the mean amount of time for transport taken as a measure of the ability and different relevant parameter regimes were investigated. For surface binding, it was observed that the ratio of the stall force to the binding force sensitivity is the main parameter that affects the mean binding time to the surface. Finally, models for the unbinding of the depolymerization motor, PilT, from the pilus base was compared between two different papers for agreements.

Contents

1 Introduction

Type IV pili are extracellular, string-like appendages that reside on the outer membrane of many Gram-positive and Gram-negative bacteria which are made of monomers called pilin that reside inside the cell membrane and are polymerized (depolymerized) by ATPases called PilB (PilT). With the activation of these ATPases, the pilus length ($\sim 1\mu m$) actively extends (retracts) and binds to different cells, surfaces, and molecules. Such functionalities of the pilus allow the cell to perform natural transformation, biofilm formation and twitching motility. (Craig et al. (2004))

1.1 Natural Transformation

The mechanisms of retraction and extension, together with their ability to bind molecules, have been topics of active research for many years and have significant applications in medicine and drug development. The ability of T4P to bind DNA enables horizontal gene transfer, allowing some pathogenic bacteria to take up DNA from their environment. (Ellison (2024)). The DNA can be bound to the T4P tip (or shaft), brought through the cell by retraction of the pilus, and enter through the cell envelope. Such a mechanism increases the genetic diversity within populations (Domingues et al. (2012)) and allows bacteria to adapt quickly to environmental changes, including exposure to antibiotics (Ellison et al. (2018)).

In this thesis, the ability of molecules to bind to the pilus and be brought to the cell was examined by simulating a pilus under competitive binding model (Koch et al. (2021)) where a molecule was placed in its area of influence. Starting from a 1D model and slowly progressing towards 2D, the mean time for the pilus to bind the object and transport it into the cell was examined for various different parameters like binding affinity(rate) of pilus tip or the size of the molecule. The simulations, then, compared with the theoretical calculations to gain insight about the influence of the parameters on the overall time period.
There exist studies that speculate on the possibility of DNA binding not only through the pilus tip but also via the shaft. Such a hypothesis is logical given that the main constituents of the pilus, pilins, have a binding affinity for DNA (Craig et al. (2004)). Also some of the minor pilins that constitude the pili also have DNA affinity and have been shown to be incorporated throughout the pilus shaft (Giltner et al. (2010), Piepenbrink et al. (2015)). This binding through the shaft is also included in the analysis of the DNA uptake, and the overall effect of length of this active region was demonstrated.

1.2 T4P Mediated Cell Adhesion

The ability of bacteria to migrate is also mediated by T4P (Burrows (2012)) in a process called twitching. The pilus tip can bind to the surface on which the cell resides and apply high levels of force—up to $100pN$ (Maier et al. (2002)). This allows some cells to move forward and backward in a manner similar to a tug-of-war. Such a mechanism enables bacteria to form biofilms, explore their environment, and exhibit collective movements under flow conditions, as Pseudomonas aeruginosa does in the urinary tract (Shen et al. (2012)).

To further understand bacterial migration and adhesion, it is important to examine the effect of dynamic interactions on binding and cell movement. The effect of the tension force generated by the pilus on retraction velocity is well known to be almost linear (Maier et al. (2004)). The influence of such tension on pilus binding was investigated in detail by Bell (1978).
In this thesis, the effect of the combination of the linear force-velocity relationship and mechanistic influences on chemical binding on the mean binding time of the pilus to the surface is analyzed both analytically and numerically. Since the binding and retraction of pili are independent, the mean binding time of the pili has a crucial effect on collective and additive force generation.

2 Biological Background

2.1 Structure of Pilus

Type IV pili (T4P) are thin, hair-like, flexible filaments composed of polymerized pilin proteins that reside on the cell membrane of many Gram-positive and Gram-negative bacteria (Craig et al. (2004)). T4P can vary their size ($\sim 1\mu m$) by the motor proteins that reside on the peripheral region of the cell membrane and also vary in numbers by the same agents that play a role in the length changes (Simsek et al. (2023)).

T4P structure has many functionalities like cell adhesion (Pelicic (2023), Denise et al. (2019)), biofilm formation (Xia (2018)), natural transformation (Ellison (2024)) and twitching motility (Burrows (2012)).

2.1.1 Pilin Structure

The pili are made out of small helical subunits called pilin, which have a length of a few microns and thickness of $50 - 80\mathring{A}$. They are stored in the inner membrane before they are polymerized into the rod-shaped pilus (Craig et al. (2004)).

Although the overall structure and amino acid sequences of pili can vary, the pili share some common features. One crucial structure of pili is the conserved N-terminal-α-helix (which is known as $\alpha 1 - N$, $\sim 85\mathring{A}$ long, 53-residue long) and a C-terminal half ($\alpha 1 - C$), which has an end that is embedded in a globular head domain (Craig et al. (2004)). This helix structure that constitutes the backbone of pili makes it robust against forces up to $100pN$ (Maier et al. (2002)). The C-terminal half is embedded in a globular domain, which is made up of three structures: a 4-stranded (5 in some type IVb) β-sheet, $\alpha\beta$-loop, and a D-loop. The globular domains of pili are responsible for surface interactions (McCallum et al. (2019)).

Type IV pili are mainly divided into two subclasses: type IVa and type IVb, considering their folding and amino acid differences. Type IVa and Type IVb pilins respectively have different leader sequence lengths: $5-6$ amino acids vs. $15-30$ amino acids; mature sequence lengths: ~ 150 amino acids vs. ~ 190 amino acids and N-methylated N-terminal residues: phenylalanine for type IVa and varying for type IVb. Species that are mentioned frequently in the references of this text, such as Neisseria gonorrhoeae and Pseudomonas aeruginosa, have Type IVa pili (Craig et al. (2004)).

2.1.2 Pilus Structure (Pilin Assembly)

T4P has a rod-like shape that can undergo a retraction or extension and change its overall length by polymerizing or depolymerizing its units. Pili are made out of smaller units called pilin, which are bound together in a hollow, right-handed helical shape with a $\sim 12\AA$ inner diameter, $\sim 52\AA$ outer diameter and $\sim 40\AA$ (4 units per turn) pitch (Folkhard et al. (1981)). (More at Sec.2.3)

2.2 Molecular Motors PilB and PilT

Polymerization and depolymerization of the pilin subunits for the pilus is achieved by two ATPases, PilB and PilT, residing inside the inner membrane and connected to the pilin monomers by an inner membrane protein called PilC. The polymerization (depolymerization) of the monomers is achieved by clockwise (counter-clockwise) rotations of PilC, caused by the motors PilB (PilT), looking from the cytoplasmic side, and this feature is responsible for the helical assembly of pilin monomers (McCallum et al. (2019)). These motor activities result in an increase (or decrease) in the pilus length and are crucial for major pilus functionalities such as natural transformation (Aas et al. (2002)) and surface colonization (Simsek et al. (2023)).

It has been shown that every pilus has only one motor connected to it at the same time (Chang et al. (2016), Maier et al. (2004)). However, the conditions of motor binding and exchanges of the motors still need to be investigated for different bacterial types. One hypothesis was made for the depolymerization motor, PilT, to have a tendency to take over when the pilus tip interacts with a surface. The pilus creates a mechanical stress that carries the information to the inner membrane to trigger the retraction (Talà et al. (2019)). Contrary to these results, it has been shown by Koch et al. (2021) that such a relationship does not exist, and that the rates (or similarly, the average times) between different motors are independent of the tension of the pilus or the pilus tip interactions. In the paper by Simsek et al. (2023), it was found that the average pilus size for the cells on different surfaces varies, where this result might be due to the retraction speed change caused by the tension and not by the rate of motor exchange.

Another dynamic that is still relevant to debate is the connection (disconnection) of PilT for the inactivation (when the pilus length reaches 0) of pilus. In the competitive binding model (see Methods), the retraction motor PilT and PilB bind to the pilus base with certain fixed rates in wild-type bacteria. Whether or not PilT remains bound when pilus length reaches zero and if it can bind again during the inactive state of the pilus was experimentally verified by (Koch et al. (2021)), and it was found that the binding of the

PilT motor to the pilus base is not affected by the pilus being present. The mean amount of time spent during the inactive state was calculated here in this thesis to make a comparison (see Results).

Although the existence of the pilus does not change the affinities of the PilT and PilB motor bindings, the force on the pilus does in fact change the retraction and extension dynamics. In a paper by Maier et al. (2004), it was shown that applying a force to the pilus can alter its retraction velocity and even change a retracting pilus to an extending one. Using mutant Neisseria gonorrhoeae with a derepressible PilT gene (MS11-600, MW4) to reduce PilT levels, it was shown that the concentration of PilT motor has a significant contribution to the reversal of pilus retraction. While for wild-type bacteria no elongation by force was detected, the mutant bacteria showed a significant frequency for retraction inversion. The pili showed a reduction of retraction velocity linearly and a stall at a maximum force (around $100pN$, Maier et al. (2002)). The retraction frequency and the pausing periods during stall forces also showed a change with the changing concentration of PilT, indicating the stochastic binding of the PilT to the pilus base. It was argued by the paper that the force unbinds the retraction motor and with enough concentration, it can be replaced by a new retraction motor. However, since the level of concentration is low, it is not backed up by a new retraction motor as quickly, so elongation can be observed. It has been noted that the elongation of the pilus after the stalling is different from that of the free extension, therefore the elongation due to the force cannot be explained by a replacement by PilB.

2.3 Natural Transformation for Type IV Filaments

T4P can be found in many bacterial species (Gram-positive or negative) and has many functionalities such as twitching motility, cell adhesion, biofilm formation, and natural transformation. In this section, more details on the natural transformation functionality will be given.

Natural transformation is the name of a special type of horizontal gene transfer that happens in bacteria where the bacterium uses active methods to internalize (uptake) extracellular DNA to use it for its activities. Such an active transport from the environment can be done by T4P filaments (Ellison (2024)), Type II secretion systems, and DNA translocase complexes (Averhoff et al. (2021), Foster et al. (2021)).

The involvement of type IVa pili in natural competence was demonstrated for various different species (Vibrio Bartlett and Azam (2005), Acinetobacter Harding et al. (2013), Neisseria Piepenbrink (2019)), whereas type IVb filaments do not show results for natural competence (Piepenbrink (2019)). It is demonstrated that the pilus can bind to DNA and during retraction, DNA proximity increases and eventually, DNA is taken inside the cell membrane Ellison et al. (2018). The binding of T4P to DNA is possible due to the difference in functionality of minor pilins (the type of pilin that constitutes the pilus in the majority is called a major pilin, whereas the other type of known pilins that also play a role in the structure of the pilus are called minor pilins) or due to a difference in structure at the shaft or tip of the pilus. The main knowledge about the overall natural competence ability was tested by mutant cells in which the major pilin (PilA, PilE) was deleted. The mutated cells lost their ability to perform natural competence (Ellison (2024)).

However, it is not always the major proteins that take the role of being a DNA receptor. For example, in the Neisseria type IV pilus system, it is a minor pilin ComP (Angelov et al. (2015)) that shows affinity to DNA and a higher ability to bind some conserved small DNA sequences (Findlay and Redfield (2009)). Just like major pilins, ComP, which resides at the tip of the pilus, has a major α-helix ($\alpha 1 - C$) followed by a loop that leads into an anti-parallel $\beta-$sheet, but ComP has no α character beyond the initial helix.

There exists a large amount of literature about the binding occurring at the tip of the pilus (van Schaik et al. (2005), Ellison (2024), Ellison et al. (2018)). Nevertheless, there exist speculations about binding occurring from the pilus shaft with the idea that minor pilins that can bind to the DNA are distributed throughout the whole body of the pilus sporadically (Giltner et al. (2010), Piepenbrink et al. (2015)).

For the process of taking DNA through the outer membrane and peptidoglycan layer, the prevailing hypothesis is that T4P pulls the DNA through the cell membrane from the secretin that T4P initiates (Maier et al. (2004), Laurenceau et al. (2013), Matthey and Blokesch (2016)). However, such a procedure is difficult to accomplish if the location of the DNA binding is through the shaft of the pilus since the radius of PilQ is somewhat similar to that of the T4P. As a solution, a different channel was proposed to internalize the DNA (Johnston et al. (2014)) next to the pilus base. Such a solution can also be used to describe non-T4P DNA internalization processes by DNA diffusing through the mentioned pore.

2.4 T4P Mediated Cell Motility

Many species of bacteria are known to show active movements and migrations in the environments that they were put into. Such abilities provide evolutionary advantages to the species by allowing them to physically explore the environment and leave certain surfaces if they need to. By actively retracting their pili and using the binding affinity of the pilus tip, T4P is a well-established apparatus for cell movements (Burrows (2012)).

The location of the pili on the cell can vary significantly across different species and even different members of the same species. The uncorrelated extension-retraction movements of the pili cause them to show a rather diffusive movement on the surfaces, considering their RMS distance (Simsek et al. (2023)). Although such a direction appears randomly without any environmental disturbances, P. aeruginosa strains can form twitching zones of radius $\sim$2cm around the point of inoculation after 20h of incubation Burrows (2012), which is considerably large compared to the cell sizes. This diffusion-like movement can be altered by a flow of liquid that fills the surrounding environment, as occurs in the urinary tract. It was shown by Shen et al. (2012) that Pseudomonas aeruginosa in the urinary tract can choose the direction of moving upstream (with a zig-zag motion) and exhibit collaborative behavior.

Not all cell movements are T4P-mediated. In the work by Simsek et al. (2023), the trade-off between active and passive adhesion for Pseudomonas aeruginosa was investigated. It was found that although for small time periods the RMS displacement shows a $\sim t$ behavior (diffusive) for the bacteria, over long time scales, it is super-diffusive, indicating that the effect of pili dominates only over large time scales.

3 Theory

3.1 Stochastic Processes

A stochastic process is a change of a variable, S, that can assign certain values to itself in a probabilistic manner. The possible values of the process form a set, which can have finitely many, uncountably many, or countably many members.

3.1.1 Definitions

The probability of having a system at states $\{s_i\}_{i=1,2,\ldots,n}$ at times $\{t_i\}_{i=1,2,\ldots,n}$ ($t_i > t_j$ if $j > i$), respectively, is defined by:

$$p(s_1, t_1; s_2, t_2, ...; s_n, t_n) := Prob\{S(t_1) = s_1; S(t_2) = s_2; ..., S(t_n) = s_n\} \tag{1}$$

For continuous states, the previous definition is slightly modified (from now on, only the continuous case will be written):

$$\begin{aligned} p(s_1, t_1; s_2, t_2, \ldots; s_n, t_n) ds_1 ds_2 \ldots ds_n := \mathrm{Prob}\big\{ & s_1 \leq S(t_1) \leq s_1 + ds_1; \\ & s_2 \leq S(t_2) \leq s_2 + ds_2; \ldots; s_n \leq S(t_n) \leq s_n + ds_n \big\}. \end{aligned} \tag{2}$$

Transition probability densities are defined as:

$$p(s_1, t_1; \ldots; s_n, t_n \mid s_{n+1}, t_{n+1}; \ldots; s_{n+m}, t_{n+m}) := \frac{p(s_1, t_1; \ldots; s_n, t_n; s_{n+1}, t_{n+1}; \ldots; s_{n+m}, t_{n+m})}{p(s_{n+1}, t_{n+1}; \ldots; s_{n+m}, t_{n+m})} \tag{3}$$

where

$$p(s_{n+1}, t_{n+1}; \ldots; s_{n+m}, t_{n+m}) = \int ds_1 ds_2 ... ds_n \; p(s_1, t_1; \ldots; s_n, t_n; s_{n+1}, t_{n+1}; \ldots; s_{n+m}, t_{n+m}) \tag{4}$$

3.1.2 Markov Property

If a probability distribution satisfies the following property:

$$p(s_1, t_1 | s_2, t_2, \ldots s_n, t_n) = p(s_1, t_1 | s_2, t_2) \tag{5}$$

for every possible choice of $\{(s_i, t_i)\}_{i=1,2\ldots n}$, then such a stochastic process is called *Markovian* or a *Markov Process*. Intuitively, this property means that the system has no memory and the future states can be determined only by knowing the information about the current state.
One important property of Markov processes is that the probability density can be written as a product of transition probabilities:

$$p(s_1,t_1;s_2,t_2;\dots;s_n,t_n) = p(s_1,t_1 \mid s_2,t_2;\dots;s_n,t_n)p(s_2,t_2;\dots;s_n,t_n) \tag{6}$$
$$= p(s_1,t_1 \mid s_2,t_2)p(s_2,t_2 \mid s_3,t_3)\dots p(s_{n-1},t_{n-1} \mid s_n,t_n)p(s_n,t_n) \tag{7}$$

Therefore, knowing the transition rates will be sufficient to understand the system. From now on, only Markovian processes will be considered.

3.1.3 Forward Master Equation

Let $p(S,t)$ be the probability of the system being in state S at time t and let $p(S,t|S',t')$ be the probability of occupying state S at time t given that the system was at state S' at time t' (transition probability). Then the following identity describes the possible ways for the system to evolve:

$$p(S,t+dt) = \sum_{S'} p(S,t+dt|S',t)p(S',t) + \Big(1 - \sum_{S'} p(S',t+dt|S,t)\Big)p(S,t) \tag{8}$$

where the first term represents the different ways for the system to evolve from the state S' to S in a short time interval dt and the second term represents the probability of starting at state S and not changing the state in the given time interval.
Defining

$$\alpha(S,S')dt := p(S,t+dt|S',t) \tag{9}$$

and substituting it inside the previous equation:

$$\partial_t p(S,t) = \sum_{S'} \alpha(S,S')p(S',t) - p(S,t)\sum_{S'} \alpha(S',S) \tag{10}$$

one finds the *Forward Master Equation (FME)*. FME can also be written with transition probabilities:

$$\partial_t p(S,t|S_\circ,t_\circ) = \sum_{S'} \alpha(S,S')p(S',t|S_\circ,t_\circ) - p(S,t|S_\circ,t_\circ)\sum_{S'} \alpha(S,S') \tag{11}$$

where $S_\circ, t_\circ$ represent any given initial condition. Defining the probability vector $(\vec{p}(t))_S := p(S,t)$, FME can be written as:

$$\partial_t \vec{p}(t) = M_f \vec{p}(t) \quad \text{where} \quad (M_f)_{SS'} = \alpha(S,S') - \delta_{SS'}\sum_{S''} \alpha(S'',S) \tag{12}$$

with the solution:

$$\vec{p}(t) = e^{M_f t}\vec{p_0} \tag{13}$$

where $\vec{p_0} = \vec{p}(t=0)$.

3.1.4 Waiting Time Analysis

The probability distribution for a realization to leave a certain state has crucial importance for simulating stochastic processes (see Sec.4.1). Here in this chapter, this probability distribution was examined and the probability of moving to any other possible states from a current state is investigated.

Suppose we have a realization at state S and let $\{S_i\}_{i=1,2,\ldots n}$ be the possible states for the realization to jump from the initial state S. Also let $\{\alpha_i\}_{i=1,2,\ldots n}$ be the transition rates, respectively. Assigning $p(S, t = 0) = 1$, one can calculate the probability of being at any other state at any other time using Eq.12 and figure out the transition probability distribution. Neglecting all the other transitions between the states except the ones that go out from S:

$$\partial_t p(S,t) = -\Big(\sum_{i=1}^{n} \alpha_i\Big) p(S,t) \quad \Rightarrow \quad p(S,t) = e^{-t\sum_{i=1}^{n}\alpha_i} \tag{14}$$

$$\partial_t p(S_i,t) = \alpha_i p(S,t) \quad \Rightarrow \quad p(S_i,t) = \frac{\alpha_i}{\sum_{i=1}^{n}\alpha_i}\Big(1 - e^{-t\sum_{i=1}^{n}\alpha_i}\Big) \tag{15}$$

Two things can be learned from this calculation: First, the probability of not changing the state, Eq.14, is exponentially decaying in time with a time constant of the sum of the transition rates, called propensity. Second, if a state change happens, the probability of going to state S_i is proportional to the corresponding transition rate α_i, since $(\partial_t p(S_i,t))/(\partial_t p(S_j,t)) = \alpha_i/\alpha_j$. By normalizing, the probability of going to state S_i in case of a state change, can be found to be $\alpha_i / \sum_{i=1}^{n} \alpha_i$.

In the case where the transition rates are not constants and are in fact time dependent, then the equations above will be slightly modified:

$$p(S,t) = e^{-\int_0^t \sum_{i=1}^{n} \alpha_i(t')dt'} \tag{16}$$

and the probability of going to another state S_i will be:

$$\alpha_i(t) / \sum_{i=1}^{n} \alpha_i(t) \tag{17}$$

3.2 First Passage Time Analysis

The question of when a certain event occurs in a certain state space for the first time is an important problem in stochastic processes. One important and frequently encountered problem is to figure out the mean time for a certain realization to leave (or pass the boundary of) or reach to a certain state space, called mean first passage time (MFPT) (Gardiner (2009), Iyer-Biswas and Zilman (2015), Polizzi et al. (2016), Kim (1958), Zhang and Melnik (2008)). In this section, the relevant methods to calculate the mean time values were analyzed.

3.2.1 For Solvable Forward Master Equations

In this section, the probability distribution for a system to reach a certain state S for the first time is analyzed for cases where the FME (Eq.12) is solvable.

To achieve it, one cannot naively check the value of $p(S,t)$, for $p(S,t)$ also includes all realizations which the system finds itself at state S before time t. To resolve this issue, one can set all the transition rates out of state S to be zero $\alpha(S',S) = 0$ for all S'. Let $P_S(t)$ be the probability of not reaching the state S until time t, then:

$$P_S(t) = \sum_{S' \neq S} p(S',t) \tag{18}$$

since now $p(S',t)$ cannot not include any instance where the system finds itself at state S before time t. If one can solve Eq.13 for absorbing boundary conditions:

$$P_S(t) = \sum_{S' \neq S} \sum_{S''} (e^{M_f t})_{S'S''} (\vec{p_0})_{S''} \tag{19}$$

where M_f is the transition rate matrix of the forward master equation, where the rates out of state S are removed.

3.2.2 For Non-Solvable FME Cases

It is not always possible to compute the exponential of a matrix when the state space is large, even with many available software packages. In such cases, there are alternative ways for one to calculate the MFPT for a state.

Suppose again one certainly starts at a state, call S_0, and calculates the MFPT to reach S_f. Again, similar to the previous discussion, one can declare S_f to be an absorbing state. Define $\tilde{M}_f$ to be the reduced FME matrix (Eq.12) where it is obtained by removing the rows and columns of the FME matrix which stores the rates going in and out of S_f. Similarly, define $\tilde{p}(t)$ to be the reduced probability vector which is obtained by removing S_f. It is true that:

$$\partial_t \vec{\tilde{p}}(t) = \tilde{M}_f \vec{\tilde{p}}(t) \quad \text{where} \quad (M_f)_{SS'} = \alpha(S,S') - \delta_{SS'} \sum_{S''} \alpha(S'',S) \tag{20}$$

where neither S nor S' is S_f. The sum on the right hand side, on the other hand, includes S_f since the rates that go to S_f are not removed. Such a system dies out as $t \to \infty$, however, by summing up the probabilities of the states (not including S_f), one can figure out the survival probability of not reaching the state S_f, call $P_{S_f}(t)$:

$$P_{S_f}(t) = \sum_{S \neq S_f} \tilde{p}_S(t) \tag{21}$$

Let $p_{S_f}(t)$ be the probability distribution of reaching the state S_f, $\int_0^t p_{S_f}(t')dt' = 1 - P_{S_f}(t)$. Then it must be that $p_{s_f} = -\partial_t \sum_{S \neq S_f} \tilde{p}_S(t)$. Calculation of MFPT ($T_{S_f}$) gives:

$$
\begin{aligned}
T_{S_f} &= -\int_0^{\infty} t\partial_t\Big(\sum_{S\neq S_f}\tilde{p}_S(t)\Big)dt \\
&= -t\Big(\sum_{S\neq S_f}\tilde{p}_S(t)\Big)\Big|_0^{\infty} + \sum_{S\neq S_f}\int_0^{\infty}\tilde{p}_S(t)dt \\
&= \vec{\tilde{1}}^T\cdot\int_0^{\infty} e^{\tilde{M}_f t}\vec{\tilde{p}}_S(t=0)dt
\end{aligned}
\tag{22}
$$

where $\vec{\tilde{1}}^T = (1,1,1,...,1)$. Using the fact that $e^{\tilde{M}_f t} = \tilde{M}_f^{-1}\frac{d}{dt}e^{\tilde{M}_f t}$, one finds:

$$
T_{S_f} = \vec{\tilde{1}}^T\cdot\tilde{M}_f^{-1}\vec{\tilde{p}}_S(t=0) \tag{23}
$$

therefore, one does not need to compute the exponential of the FME, but only its inverse, which is much more manageable computationally.

3.2.3 Kramers Method for the Mean First Passage Time

On the question of mean times to escape from a certain state space, one method that is widely known is the Kramers method (Kramers (1940), Hänggi et al. (1990)). In this thesis, Kramers method was used to calculate the mean amount of time needed for the pilus tip to reach a certain length, $L_{threshold}$, and come back to the inactive state.

Kramers method for single flux can be summarized as such: Let G be a subregion of the whole state space and assume x_0 is the only entrance to G. Kramers' rate is given by:

$$
k_G(x_0) := \frac{J_{x_0}}{\int_G P(x)dx} \tag{24}
$$

where J_{x_0} is the equilibrium probability flux to the region G and $P(x)$ is the equilibrium probability density. It can be proved (Reimann et al. (1999)) that the mean first escape time from G for a realization that enters the region via x_0 is given by:

$$
T_G(x_0) = \frac{1}{k_G(x_0)} = \frac{\int_G P(x)dx}{J_{x_0}} \tag{25}
$$

For a generalization to multiple entrances to the region G, the method provides a relation between the MFPT values for realizations that enters the region from different states (Reimann et al. (1999)):

$$
\frac{1}{k_G[J(x)]} := T_G([J(x)]) = \frac{\int_G J(x)T_G(x)dx}{\int_G J(x)dx} = \frac{\int_G P(x)dx}{\int_G J(x)dx} \tag{26}
$$

where the $T_G[J(x)]$ is the mean first escape time from G for a flux distribution of enterances $J(x)$. As expected, $T_G([J(x)])$ is a weighted average of the escaped times from different enterances, $T_G(x)$, since the larger the $J(x)$ value, the more instances there are in the sum of escape times for the realizations that enter into the region G.

A proof (similar to the one given at Reimann et al. (1999)) for the generalized case (Eq. 26) for finitely many source terms $J(x) = J_i\delta(x - x_i)$ can be given as such:
Suppose there are N many sources $\{x_n\}_{n=1,2,...,N}$ through the region G with constant fluxes $\{J_n\}_{n=1,2,...,N}$ respectively. Suppose one takes M many realizations that starts from each of the source states x_i at time $t = 0$. Assuming absorbing boundary conditions on G, one can define:

$$\rho_n(t) = \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} \Theta(\tau_i^n - t) \quad \text{for each} \quad n = 1, 2, \ldots, N \tag{27}$$

where $\rho_n(t)$ is the fraction of realizations that start at x_n and has left the region G at time t and τ_i^n is the time it takes for i^{th} realization that starts at x_n to leave the region G for the first time. Then, the mean escape time for realizations that enter the region through x_n is given by:

$$T_G(x_n) = \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} \tau_i^n \tag{28}$$

At any time $t > 0$, the total amount of probability in region G can be described as:

$$\int_G P(x,t)dx = \sum_{n=1}^{N} \int_0^t J_n \rho_n(t-s)ds \tag{29}$$

Making the transformation $t' = t - s$:

$$\int_G P(x,t)dx = \sum_{n=1}^{N} J_n \int_0^t \rho_n(t')dt' \tag{30}$$

Using Eq.27:

$$\int_G P(x,t)dx = \sum_{n=1}^{N} J_n \left(\lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} \int_0^t \Theta(\tau_i^n - t')dt' \right) \tag{31}$$

Large t values bring the system into an equilibrium state, namely $P(x,t) \to P(x)$. Also the integral on the right-hand side simplifies: $\int_0^t \Theta(\tau_i^n - t')dt' \to \tau_i^n$. One gets:

$$\begin{aligned} \int_G P(x)dx &= \sum_{n=1}^{N} J_n \Big(\lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} \tau_i^n \Big) \\ &= \sum_{n=1}^{N} J_n T_G(x_n) \end{aligned} \tag{32}$$

To find the mean first escape time from G for a flux distribution $\{J_n\}_{n=1,2,...,N}$, call it $T_G(\{J_n\})$, one can use the mean escape times for individual starting states $T_G(x_n)$. $T_G(\{J_n\})$ must be a weighted average of $T_G(x_n)$ values with the weights J_n because the higher the magnitude of J_n the higher number of realizations starting from the state x_n are counted in the overall average of escape times. Therefore, by using Eq.32:

$$T_G(\{J_n\}) = \frac{\sum_{n=1}^{N} J_n T_G(x_n)}{\sum_{n=1}^{N} J_n} = \frac{\int_G P(x)dx}{\sum_{n=1}^{N} J_n} \tag{33}$$

which is identical to what is given by Eq.26.

3.3 Analytical Model for Pilus Kinetics

Two parameters were used to describe the state of the pilus: The motor state which can be *inactive*, *retracting*, *extending* or *idle* and a continuous parameter, length. The length changes depending on the state of the pilus: the pilus extends (retracts) with the rate v_e (v_r) in the state *extending*(*retracting*). During the *idle* state, the length of the pilus does not change. The *inactive* state is the state where the pilus length is equal to zero, i.e. the pilus does not exist. The rates and changes between the states are given in the *Methods* section.

For a given time t, the probability distribution of finding the pilus at a certain state and length l is denoted by: $P_\circ(l,t), P_e(l,t), P_r(l,t)$ for *idle*, *extending* and *retracting* states, respectively. The probability of finding the pilus in the *inactive* state is notated as $P_{inactive}(t)$.

3.3.1 Steady State

To find the master equation regarding a state, we start by the *retracting* state. We add the probabilities of possible ways to end up in that given state:

$$P_r(l, t+dt) = k_{0r} dt P_0(l,t) + (1 - k_{r0}dt) P_r(l + v_r dt, t) \tag{34}$$

where the first term represents the probability of state to change from *idle* to *retract* and the second term is staying at the same state but retracting from $l + v_r t$ to l. Expanding the second term for l and canceling high powers of dt:

$$P_r(l, t+dt) = k_{0r} dt P_0(l,t) + (1 - k_{r0}dt)(P_r(l,t) + P_r'(l,t) v_r dt)$$

$$P_r(l, t+dt) = k_{0r} dt P_0(l,t) + P_r(l,t) + P_r'(l,t) v_r dt - k_{r0} dt P_r(l,t)$$

$$\partial_t P_r(l,t) = k_{0r} P_0(l,t) - k_{r0} P_r(l,t) + v_r P_r'(l,t) \tag{35}$$

Idle and *extending* state equations can be found similarly.

For the *inactive* state, we follow the same approach but with some differences:

$$P_{inactive}(t+dt) = (1 - k_{on}dt) P_{inactive}(t) + (1 - k_{r0}dt) \int_0^{v_r dt} P_r(l', t) dl'$$

Where the second term on RHS represents the probability of shrinking to size zero. Expanding the terms will result:

$$\partial_t P_{inactive}(t) = -k_{on} P_{inactive}(t) + v_r P_r^(0, t) \tag{36}$$

As $t \to \infty$, the probability distributions will reach the equilibrium distributions: $P_i(l,t) \to P_i(l)$ and $P_{inactive}(t) \to P_{inactive}$ together with $\partial_t P_i(l,t) \to 0$ Using the equations we found previously, the equilibrium equations regarding the distributions are given by:

$$0 = -(k_{0r} + k_{0e})P_\circ(l) + k_{r0}P_r(l) + k_{e0}P_e(l) \tag{37}$$

$$0 = -k_{e0}P_e(l) + k_{0e}P_\circ(l) - v_e P'_e(l) \tag{38}$$

$$0 = -k_{r0}P_r(l) + k_{0r}P_\circ(l) + v_r P'_r(l) \tag{39}$$

$$P_{inactive} = \frac{v_r}{k_{on}} P_r(l)\Big|_{l=0} \tag{40}$$

Since, for any l, the region $inactive + (0, l)$ is bounded, the equilibrium net flux through it must be zero:

$$P_r(l)v_r = P_e(l)v_e \tag{41}$$

It is also useful to define the total probability function for equilibrium:

$$P(l) := P_\circ(l) + P_r(l) + P_e(l) \tag{42}$$

which satisfies the normalization condition:

$$P_{inactive} + \int_0^\infty P(l)dl = 1 \tag{43}$$

Solutions for these equations can be found by using $P_i(l) = C_i e^{-\kappa l}$ with constants C_i and κ. Substituting this to the steady state equations provides 4 equations with 4 unknowns ($P_{inactive}$ is related to $P_r(l=0)$ by Eq.40). Solving these equations provides:

$$P_{\text{inactive}} = \frac{k_{0e}k_{r0}v_e - k_{0r}k_{e0}v_r}{v_e k_{r0}(k_{0e} - k_{on}) - k_{e0}v_r(k_{0r} + k_{on}) - k_{0e}k_{on}(v_e + v_r) - k_{0r}k_{on}(v_e + v_r)} \tag{44}$$

$$P_\circ(l) = e^{-\kappa l}\Big(\frac{k_{on}P_{\text{inactive}}(k_{e0}v_r + k_{r0}v_e)}{v_e v_r(k_{0e} + k_{0r})}\Big) \tag{45}$$

$$P_e(l) = e^{-\kappa l}\Big(\frac{k_{on}P_{\text{inactive}}}{v_e}\Big) \tag{46}$$

$$P_r(l) = e^{-\kappa l}\Big(\frac{k_{on}P_{\text{inactive}}}{v_r}\Big) \tag{47}$$

3.4 Mean Uptake Time for a Spherical Object in High Binding Rate Limit

The ability of the pilus to uptake DNA was determined by the mean uptake time required for the pilus to transport the object from the initial position to the pilus base. The calculation of this time interval requires solving the mean first passage time problem for the pilus equations. All the theoretical calculations were done in $k_{binding} \to \infty$ and $k_{unbinding} = 0$ limits.

In order to get the mean time required for the pilus to uptake a spherical object in the high binding limit ($k_{bind} \to \infty$) and low unbinding rate ($k_{unbind} = 0$), it is only necessary for one to calculate the time for pilus tip to cross a certain length threshold, $L_{threshold}$, (which makes the tip touch the object) for the first time and then come back. The tip will immediately bind to the object and does not get unbind during the transport.

In this section, both 1D and 2D models for this limit case was examined by varying the object distance. The state space for the pilus will be denoted as $[0, \infty)$, where 0 means pilus is in the *inactive* state and any other length $l > 0$ denotes pilus tip to be l units away from pilus base with any of the motor states: *retracting*,*idle* or *extending*.

3.4.1 1D Case

In 1D case, the length that pilus needs to reach is given by $L = d_{object} - R$, where d_{object} is the object's distance and R is it's radius. Since it's 1D and the pilus tip immediately binds, the geometry of the object does not matter and R can be ignored. The time total required to reach length L for the first time and then coming back will be denoted as $T(L)$.

Define the following times for the pilus tip movement:
$\tau(L)$: The average time for pilus tip to start from $l = 0$ (in *inactive* state) and pass through $l = L$ for the first time

ξ: The average time for pilus tip to start from $l = L$ in the *extending* state, wandering in $l > L$ region and cross $l = L$ from the right for the first time in the *retracting* state (it will be shown later in the text that ξ is independent of L)

$\pi(L)$: The average time for pilus tip to start from $l = L$ in the *retracting* state and reach $l = 0$ for the first time
Since any realization that uptakes the object has to take these 3 steps, it must be that $T(L) = \tau(L) + \xi + \pi(L)$, namely, the mean uptake time is equal to the sum of three time values defined above.

Also, let $\Gamma(L)$ be the average time interval for pilus to enter the region $[0, L)$ from right (in *retracting* state) and leave it again with the same length (in *extending* state). This time interval can easily be calculated by the Kramers method since the region $[0, L)$ has only one entry point (see section 3.2.3), using Eq.25:

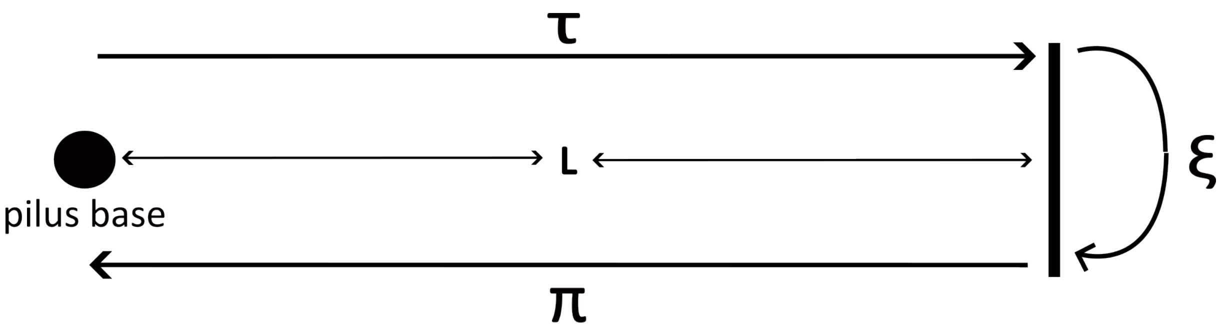

Figure 1: The total uptake time for the 1D case is divided into three regions τ, ξ and π (refer to the text for the definitions)

$$\Gamma(L) = \frac{P_{inactive} + \int_0^L P(l)dl}{P_r(L)v_r} \tag{48}$$

The value of $\Gamma(L)$ includes the instances where the pilus starts from length L in the retract position and reaches length L again without going to the pilus base. It can be argued that the contributions to the mean time coming from those instances that do not reach to the pilus base is negligible: If one checks out the state change rates for length states (see *Appendix*), the mean time for pilus to stay in *retract* state, $1/k_{r0}$ is larger than the time it takes for pilus to reach the pilus base for reasonable initial length values: $1/k_{r0} > L/v_r$. Moreover, even if the pilus goes to *idle* state, it is more likely that the pilus will go back to the *retract* state: $k_{0r} > k_{0e}$. In the light of these observations, one can assume that the pilus does not go to the *extend* state twice during it's course. Therefore, the probability (distribution) of pilus to stop in the time interval $(t, t+dt)$, go into the *idle* and *extend* states respectively and then stay at *extend* until it reaches the initial length L, can be calculated by:

$$p(t) = (k_{r0}e^{-k_{r0}t})\Big(\frac{k_{0e}}{k_{0e}+k_{0r}}\Big)(e^{-(k_{e0}v_r/v_e)t}) \tag{49}$$

where the first term is the probability of going to idle state in $(t, t+dt)$, the second term is the probability of going to *extend* state once *idle* state is on and the third term is going back to the initial length without stopping. Integrating Eq.49, one gets the fraction of instances that returns to initial length before reaching the pilus base (see Fig.2 for plot and see Fig.22 for simulation results):

$$\int_0^{L/v_r} p(t)dt = \Big(\frac{k_{r0}k_{0e}}{k_{0e}+k_{0r}}\Big)\Big(\frac{1}{k_{r0}+\frac{k_{e0}v_r}{v_e}}\Big)\Big(1 - e^{-\Big(\frac{k_{r0}}{v_r}+\frac{k_{e0}}{v_e}\Big)L}\Big) \tag{50}$$

The upper limit for integration ensures that only the realizations that do not reach the pilus base are considered.

The previous argument implies $\Gamma(L) = \tau(L) + \pi(L)$ and therefore: $\Gamma(L) + \xi = T(L)$. However, the value of ξ can also be found by Eq.25:

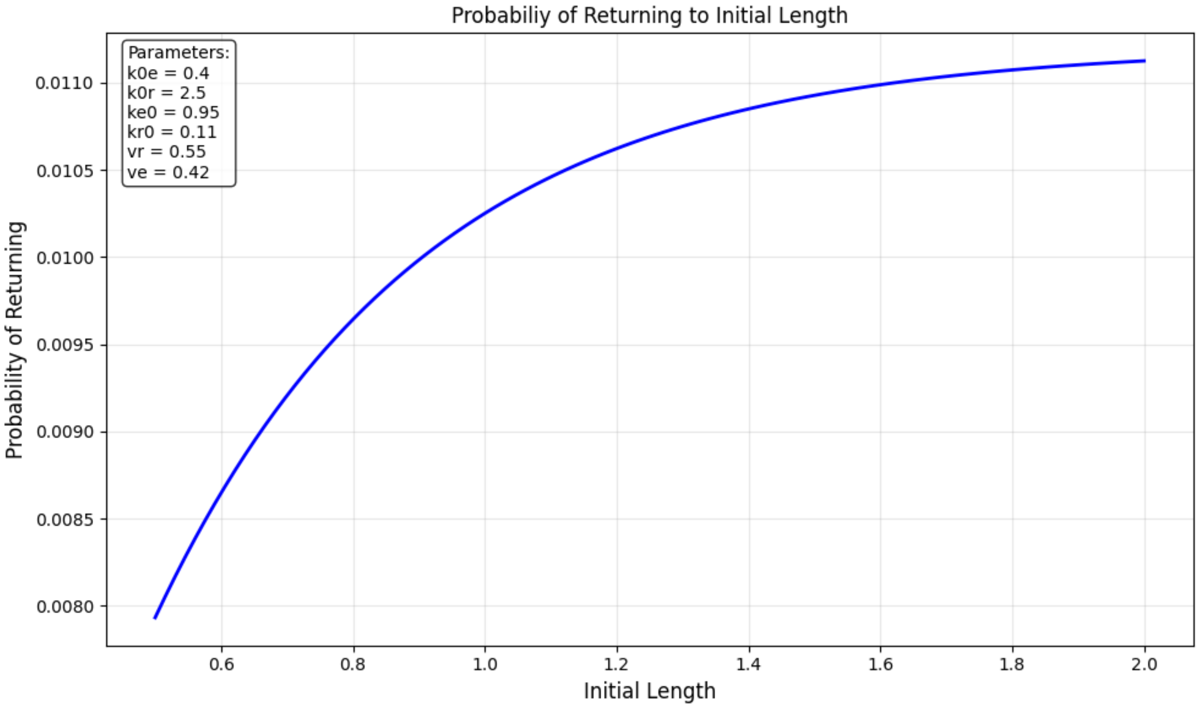

Figure 2: The probability of a pilus to start at a certain initial length at *retract* state and before going to pilus base, stop and extend back to the initial length (Eq.50). For length values around the average pilus length, only a small fraction returns back. This justifies the usage of Kramers' method for single flux (Eq.25) without much biased error. The parameters used to calculate this result is choosen by the parameters used in other simulations used this thesis. (See Fig.22 for simulation results.)

$$\xi = \frac{\int_L^\infty P(l)dl}{P_e(L)v_e} \tag{51}$$

which in the end gives:

$$T(L) = \frac{1}{P_e(L)v_e} \tag{52}$$

by using Eq.41 and 43.

3.4.2 2D Case

In the 2D case, two connections can be made to the 1D result (Eq.52): First, there will be an inclusion of the geometry of the object. Since it is the high binding rate limit, it is only the distance from the surface of the object to the origin that matters and not the bulk of the object. By using the result from Eq.52, one can find the average time it takes to reach the object surface to be:

$$T(\xi L) = \frac{1}{P_e(\xi L)v_e} \tag{53}$$

where ξ is given by Eq.58.

Secondly, since total angle of incidence ($2\phi = 2sin^{-1}(R/d)$, see Fig.3) of the object is smaller than the genesis angle (Δ, see Fig.6), not all pilus activations are in the right angle to meet with the object even if they happen to reach significant lengths. The overall probability for the pili to have a direction towards the object is given by:

$$\alpha = \frac{2\phi}{\Delta} \tag{54}$$

by the fact that the angles are choosen uniformly between $(0, \Delta)$.

To include these effects, one can assume that the number of different pilus activations is high enough so that the total time scales proportionally:

$$T(\xi L) = \left(\frac{1}{\alpha}\right)\frac{1}{P_e(\xi L)v_e} \tag{55}$$

as expected, $2\phi \to \Delta$ makes the contribution from the angle of incidence to diminish: $1/\alpha \to 1$.

3.5 Geometric Analysis

Using a circular object to analyze the mean uptake time requires geometric calculations. In this section, the necessary geometric analysis was made.
By the triangle that forms in Fig.3:

$$R^2 = x^2(\theta) + d^2 - 2x(\theta)d\cos(\theta) \tag{56}$$

solving for $x(\theta)$:

$$\frac{x(\theta)}{d} = \cos(\theta) - \sqrt{\cos^2(\theta) - (1-(R/d)^2)} = \cos(\theta) - \sqrt{\cos^2(\theta) - \cos^2(\phi)} \tag{57}$$

where $\phi = \sin^{-1}(R/d)$. Then averaging on $(-\phi, \phi)$ gives:

$$\xi := \frac{x(\theta)}{d} = \frac{2R}{d}(1 - E(\phi, \csc^2(\phi))) \tag{58}$$

where $E(\phi, k) = \int_0^\phi \sqrt{1-k^2\sin^2(z)}dz$ is the elliptic integral of the second kind.
The values of $\frac{\langle x\rangle}{d}$ found numerically, using Mathematica (1999).

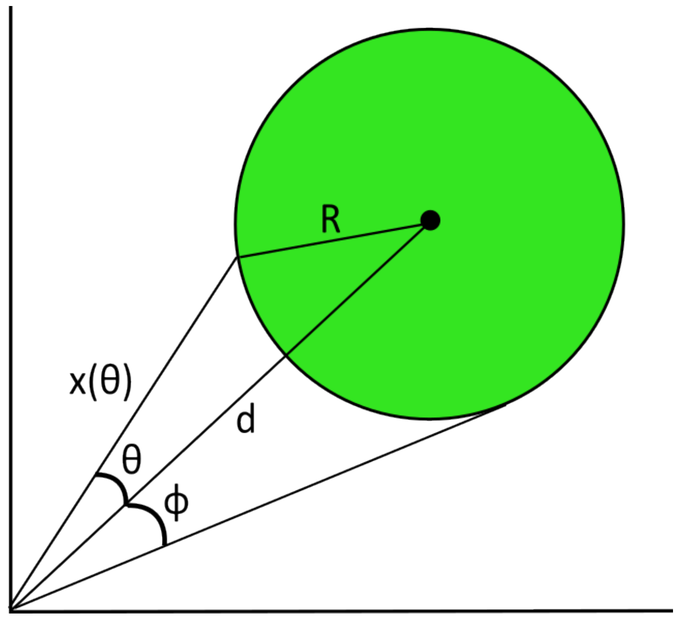

Figure 3: For $k_{bind} \to \infty$ and $k_{unbind} = 0$ cases, only the distance from the object surface to the origin matters. To ease the calculations, the mean distance from origin to the object surface $\langle x \rangle$ was calculated for $\theta \in (-\phi, \phi)$

4 Methods

4.1 Gillespie Algorithm

To simulate the state changes, the Gillespie Algorithm was employed. The Gillespie Algorithm provides a way to simulate realizations of stochastic processes in real time, given that the transition rates between states are known.

For a realization in a current state, the probability distribution of changing the state with respect to time is given by Eq.14 (by differentiating):

$$p(t) = \frac{dW(t)}{dt} = \frac{1 - S(t)}{dt} = \alpha e^{-\alpha t} \tag{59}$$

where $S(t) := 1 - W(t)$ is the probability of not changing the state in the time interval $(0, t)$ and α is called the propensity (the sum of all possible transition rates from the system's current state).

The realizations were iterated with small discrete time steps, dt. At each time step the probability of state change was calculated using Eq.59:

$$\text{Prob}(0 < T < dt) = W(dt) = 1 - e^{-\alpha dt} \tag{60}$$

where T is the time it takes for state to change and $\text{Prob}(0 < T < dt)$ is the probability that state change has happened. To determine if a state change occurs, a random variable $r_1 \in [0, 1]$ was chosen uniformly. Since r_1 is uniformly chosen, checking the condition

$r_1 < 1 - e^{-\alpha dt}$ correctly determines the probability of changing the state.

If a state change occurs, the probability of transitioning to one of the possible states is proportional to the transition rates (Eq.17). Let $\{p_i\}_{i=1,2,3,\ldots,N}$ be the probabilites of going to possible states ($p_i = k_i/\alpha$ with k_i being the transition rate). Using the normalization condition $\sum_{i=1}^{N} p_i = 1$, one can partition the interval $(0,1)$ into N segments:

$$[0,1] = [0,p1] \cup [p1, p1+p2] \cup \ldots \cup \Big[\sum_{i=1}^{N-1} p_i, 1\Big] \tag{61}$$

Since these subintervals have lengths proportional to the probabilities, a random variable $r_2 \in [0,1]$ can be drawn uniformly. By determining which subinterval it falls into, a state can be chosen with the given probability weights.

Overall, the algorithm can be summarized as follows:

- **Determine if a state change happens**
  - Calculate $p = 1 - e^{-\alpha dt}$
  - Pick $r_1 \in (0,1)$ uniformly
  - If $r_1 \geq p$ no state change happens
  - If not continue with other steps
- **Determine the new state**
  - Calculate $p_i = k_i/\alpha$
  - Pick $r_2 \in (0,1)$ uniformly
  - Determine which subinterval r_2 falls into in Eq.61
  - Pick the corresponding state
- **Change the state**

4.2 Pilus Kinetics

The pilus kinetics were implemented using a competitive binding model (Simsek et al. (2023)), where *PilB* (extension motor) and *PilT* (retraction motor) take turns affecting the length of the pilus, with no more than one motor binding to the pilus base at any time. For any motor to bind to the pilus base, the pilus base needs to be empty, therefore a *retraction* can not be followed *extension* immediately or vice versa. Instead, the motor that is already bound to the pilus base must first unbind, and the pilus base must be empty. Only then is it possible for other motors to bind.

There are two main types of parameters used to describe the pilus: The motor state, which can be *inactive* (pilus length is zero), *retracting* (represents PilT binding), *extending* (represents PilB binding) or *idle* (represents no binding by any motors) and a

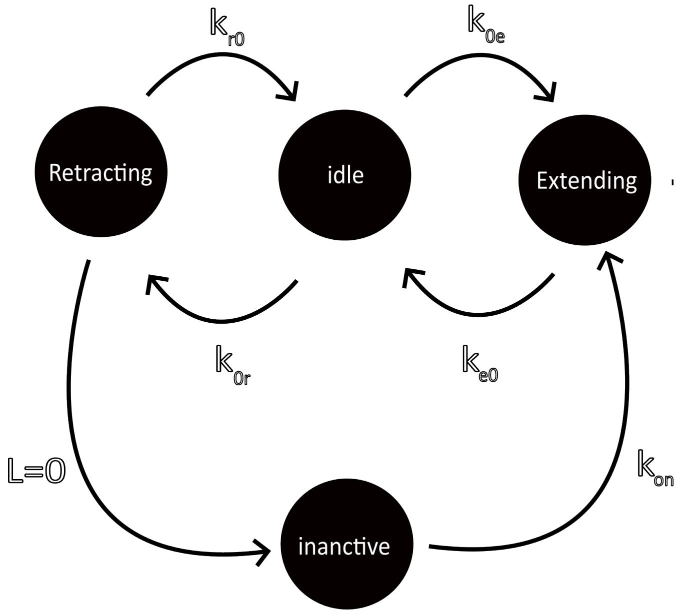

Figure 4: Rates between different motor states. The inactive state can only be reached if the pilus length diminishes.

continuous parameter, length, which takes its values from the interval $[0, \infty)$.

The length variable's value changes depending on the motor state of the pilus: the pilus extends (retracts) with a rate v_e (v_r) in the motor state *extending* (*retracting*). During the *idle* state the length of the pilus does not change:

$$\frac{dl}{dt} = \begin{cases} v_e, & \text{if } \textit{extending} \\ -v_r, & \text{if } \textit{retracting} \\ 0, & \text{if } \textit{idle} \end{cases} \tag{62}$$

The motor states change stochastically (using the Gillespie Algorithm, see 4.1) between the states *retracting*, *extending*, and *idle*. From the *idle* state, the possible states are *retracting* and *extending* with the rates k_{0r} and k_{0e}, respectively. From *extending*, the only possible state is *idle*, with the rate k_{e0} and from *retracting* it is again the idle state with the rate k_{r0} and *inactive* state if the pilus length reaches to zero.

If the length of the pilus reaches zero during the *retracting* state, the motor state changes to *inactive*, where the pilus remains at zero length.In order for the pilus to become active, the *inactive* state has to change to the only state that it can, the *extending* state, with the rate k_{on}. Then, the pilus normally continues to change its motor states as described above see (Fig. 4).

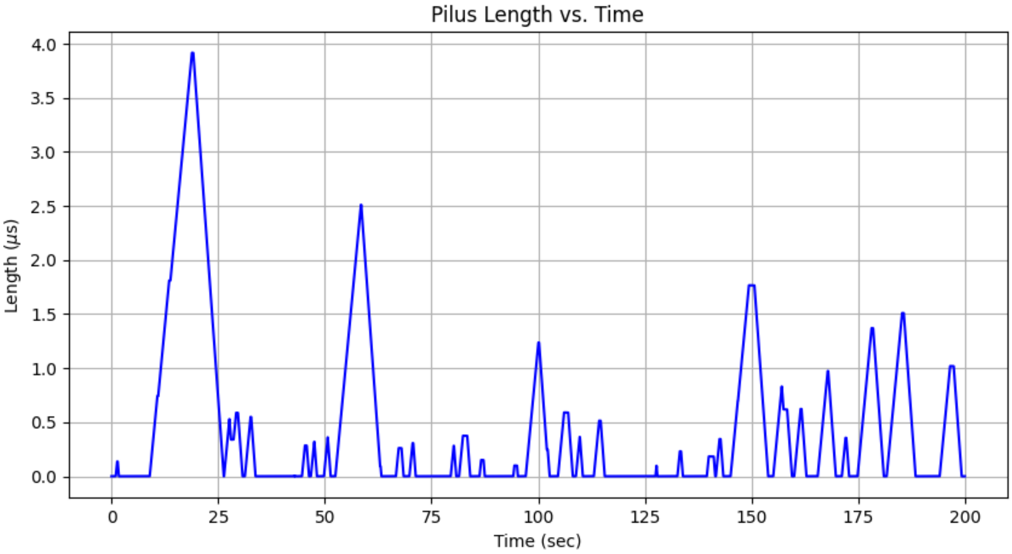

Figure 5: One example realization to the competitive binding model of the pilus length. Notice that it is not very likely for pilus tip to stop and change it's direction during the activation cycles. See Table1 in *Appendix* for the details of the simulation.

4.3 Modeling DNA Uptake

4.3.1 2D Setup

The ability of DNA uptake of the pilus was determined by placing a spherical object in front of it in 2D setting (see Fig.6) and observing the mean uptake time namely the time it takes for the pilus to grab the object and transport it back to the cell.

The pilus gets active and inactive many times the before the uptake completes. Each time the pilus becomes active, its angle is chosen randomly from the interval $(0, \Delta)$ in a uniform manner, and it maintains that angle until the next inactivation. Δ is called *genesis angle* and choosen to be $45°$ for the simulations (Simsek et al. (2023)).

4.3.2 Binding Dynamics

The condition for binding is determined by a two-state stochastic process, with states *bound* and *unbound* (see Fig.7).
The pilus is in the *unbound* state when it first becomes active. The value of k_{bind} is non-zero only when the pilus tip (or the active region, see Sec. 4.3.3) is inside the region bounded by the object:
At each time step, the simulation checks if the pilus tip (or active region) is inside the object boundaries, if so, it uses Gillespie Algorithm (see sec.4.1) in the 2-state bounding process to check whether the pilus gets binded to the object (see Fig.8). If this occurs, the location of the object is updated by the same distance and direction as the pilus tip, as long as the *bound* state is active. This procedure ensures that the pilus tip does

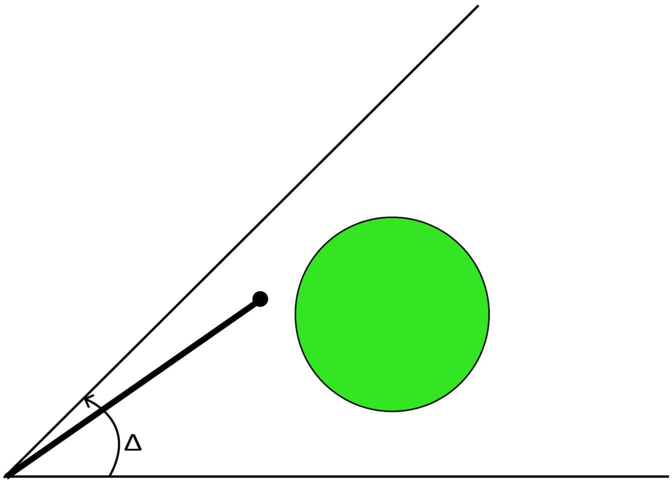

Figure 6: 2D setup for the calculation of uptake time. Each time the pilus gets activated a random angle was choosen from $(0, \Delta)$. The pilus maintains its chosen angle until the next inactivation.

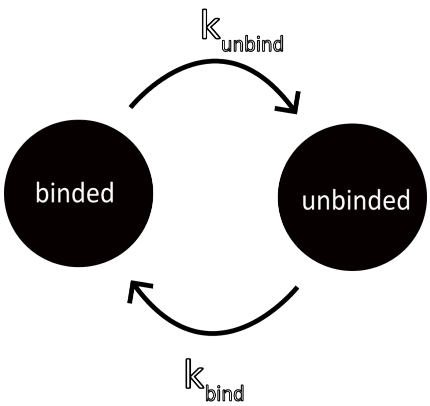

Figure 7: Rates between different bind states

not leave the space surrounded by the object boundary unless the *unbound* state is active.

When the pilus tip is binded to the object, the object gets carrried to the pilus base and when the object is finally at the pilus base the simulaion stops and the total time for the object to reach to the pilus base was recorded.

4.3.3 Active Region

In order to simulate the ability of pilus shaft to bind to the object was achieved by also allowing the pilus shaft to activate binding ($k_{bind} > 0$) by being inside the object region (see Fig.9). The region of the pilus which makes $k_{bind} > 0$ is called the *active region* and it's length is always measured from the top of the pilus. For each simulation, the length of the active region stays constant.

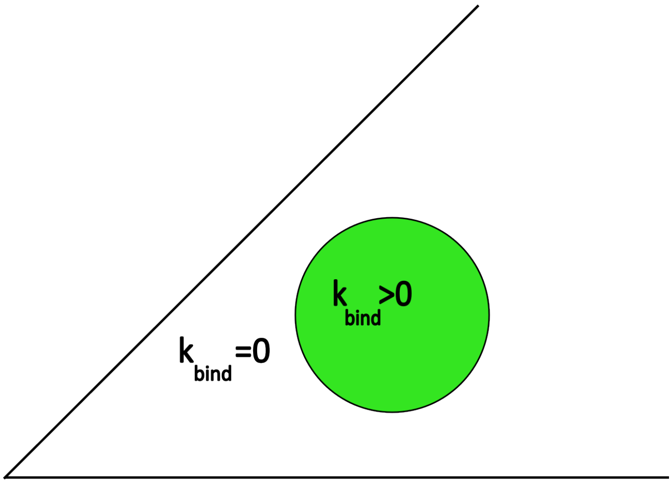

Figure 8: k_{bind} is non-zero only if the pilus tip (or the active region) is inside the region bounded by the object

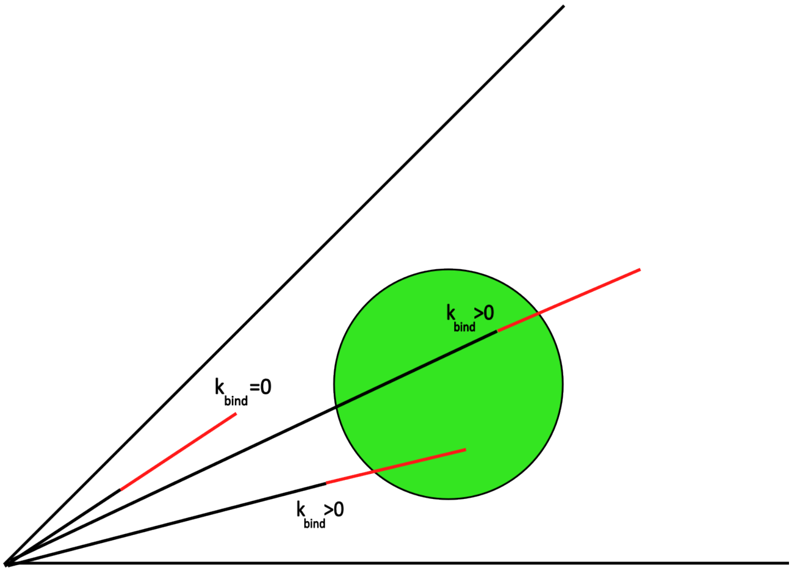

Figure 9: The ability of pilus to bind with a region at the tip (not just the tip) was simulated. k_{bind} is non-zero only if active region (shown here with red) intersects with the object.

4.4 Surface Adhesion

The ability of T4P to bind to a given organic or glass surface was investigated in 1D setting. On top of the pilus kinetics mentioned in this section, a binding similar to that of the pilus-DNA binding (Sec.4.3.2) was employed, however, this time the binding is not limited to a certain region of the space. Binding and unbinding rate constants, k_{bind} and k_{unbind}, have their non-zero values throughout the whole region.

When the pilus tip is attached to a surface (namely, in *bind* state), the size of the pilus does not change, even though *retract* or *extend* states are on. On one hand, the extension of the pilus while the pilus is attached to the surface causes it to buckle up and does not alter the dynamics of motion. On the other hand, retraction of the pilus creates a tension on the pilus body which was modelled like a spring:

$$T(\Delta L) = max(0, \kappa\Delta L) \tag{63}$$

where κ is the spring constant and ΔL is the length difference that is defined as the distance between the pilus tip and the pilus base in the presence of binding, compared to the same distance if the binding had not occurred (see Fig.10). The *max()* function was added to ensure that only the retraction causes a tension and not the extension.

The existence of the tension on the pilus body causes a reduction on the retract velocity. The relationship between the tension force and the retract velocity was modelled linearly (Clausen et al. (2009)):

$$v_r = v_{r0}\Big(1 - \frac{T(\Delta L)}{F_{stall}}\Big) \tag{64}$$

where v_{r0} is the default retraction speed, $T(\Delta L)$ is the tension on the pilus (Eq. 63) and F_{stall} is a phenomenological constant.

When unbinding occurs, the pilus tip immediately returns to the length it would have had if it had continued extending or retracting normally without binding to the surface (see Fig.10c).

The tension on the pilus causes the pilus to unbind easily, therefore a dependence on tension was set for k_{unbind}:

$$k_{unbind} = k^0_{unbind} e^{T(\Delta L)/F_{sens}} \tag{65}$$

where k^0_{unbind} is the default unbinding rate, F_{sens} is a phenomenological constant, and $T(\Delta L)$ is the tension on the pilus (Bell (1978)).

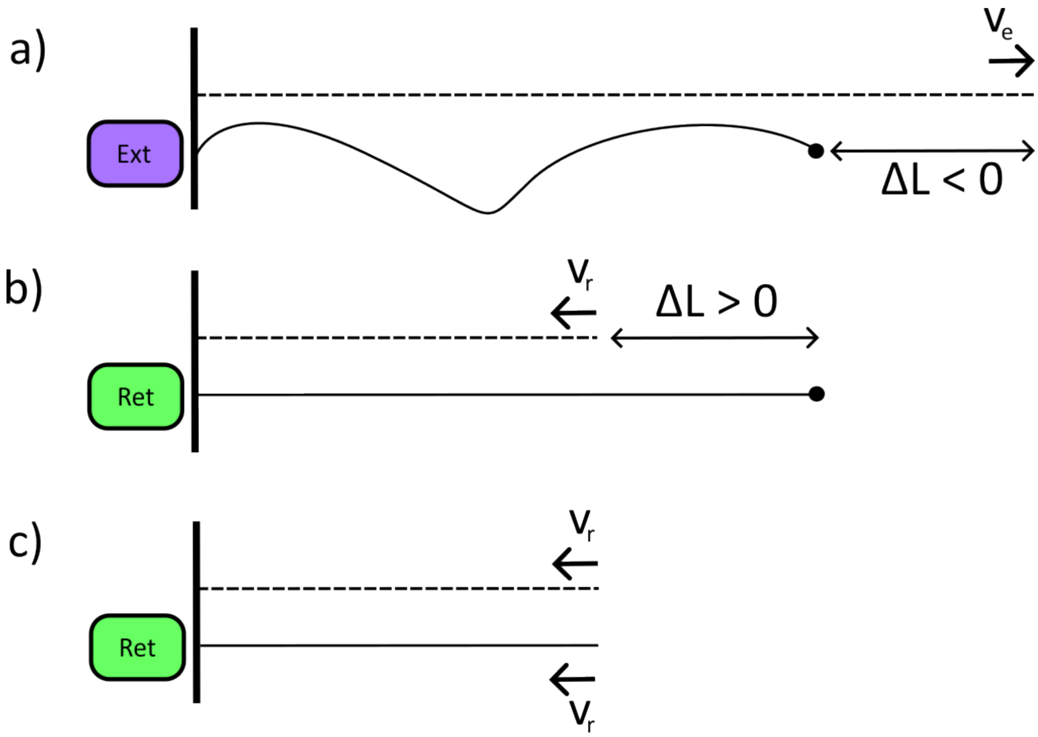

Figure 10: States of the pilus during the binding to the surface. Dashed lines indicate the length as if the binding had not occurred. The dot at the tip represents the binding. (a) The pilus extends during the binding, it buckles up and does not create a tension. (b) The pilus retrieves during the binding, creates a tension on the pilus body. (c) When unbinding occurs, the real length immediately jumps to the length as if the binding had not occcured.

5 Results

The ability of the pilus to bind to DNA from the environment was scrutinized using a competitive motor model. The ability of the pilus to uptake DNA from the environment was quantitatively investigated by measuring the mean uptake time required to transport DNA to the pilus base.

5.1 Spherical Object Uptake in High Binding and Low Unbinding Rate

The ability of the pilus to uptake a spherical object was examined under conditions of high binding ($k_{binding} \to \infty$) and a low unbinding rate ($k_{unbinding} = 0$). This regime of binding parameters causes the pilus tip to bind to the object immediately, preventing unbinding during the transportation of the object to the pilus base. To quantify, the mean amount of time for pilus to start from *inactive* state, bind the object and bring it back to the pilus base was examined. The analyses were conducted for both 1D and 2D cases.

5.1.1 1D Case

For the 1D case, the spherical object occupies an interval: $(d_{object} - R, d_{object} + R)$. The pilus tip only needs to reach the length $L = d_{object} - R$ to bind to the object. Since the pilus tip immediately binds and does not unbind, the geometry of the object becomes redundant, allowing R to be ignored. The time required for the pilus tip to start from the *inactive* state, reach a certain length $L := d_{object}$ for the first time, and then return to $L = 0$ was simulated using the competitive binding model of pilus kinetics (see Sec.4.2) with varying L values.

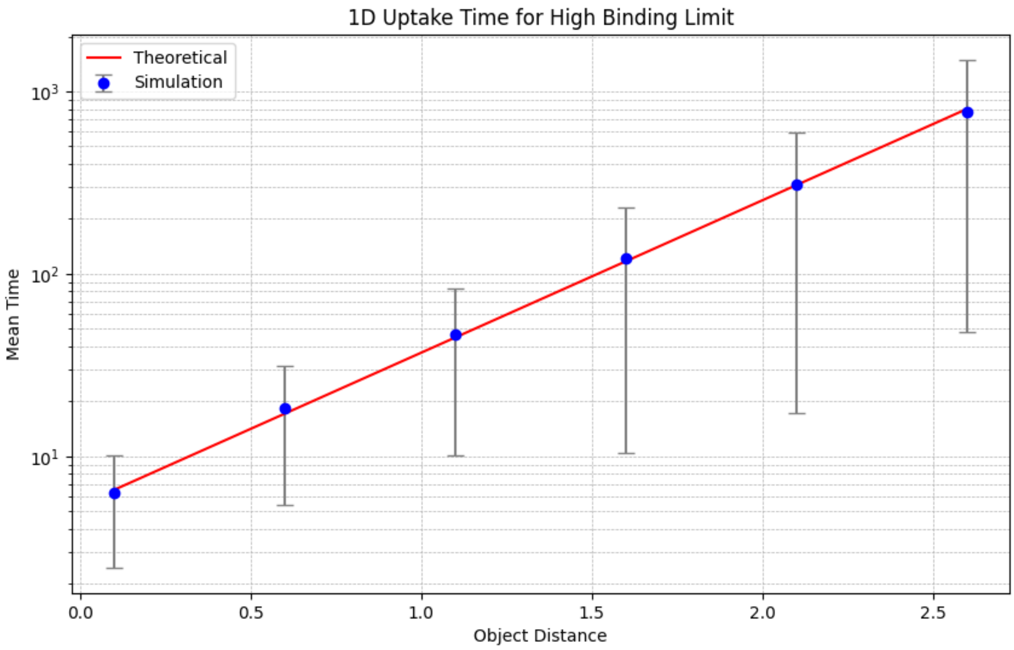

Figure 11: 1D problem with $k_{binding} \to \infty$ and $k_{unbinding} = 0$ limits. Simulation values were found by averaging the total times. The error bars represent the standard deviations. The theoretical value was determined by calculating the mean passage time for the pilus length to reach d_{object} (see Sec.3.4.1). See Table2 in *Appendix* for the details of the simulation.

$$T_{1D}(L) = (P_e(L)v_e)^{-1} \tag{66}$$

5.1.2 2D Case

For the 2D case, the object occupies a spherical region centered at $(d_{object}, \Delta/2)$ in polar coordinates (where Δ is the genesis angle, see Fig.6) with a radius of R. The time required for the pilus tip to start from the *inactive* state, reach the region occupied by the object for the first time, and then return to $L = 0$ was simulated using the competitive binding model of pilus kinetics (see Sec.4.2) with varying R values.

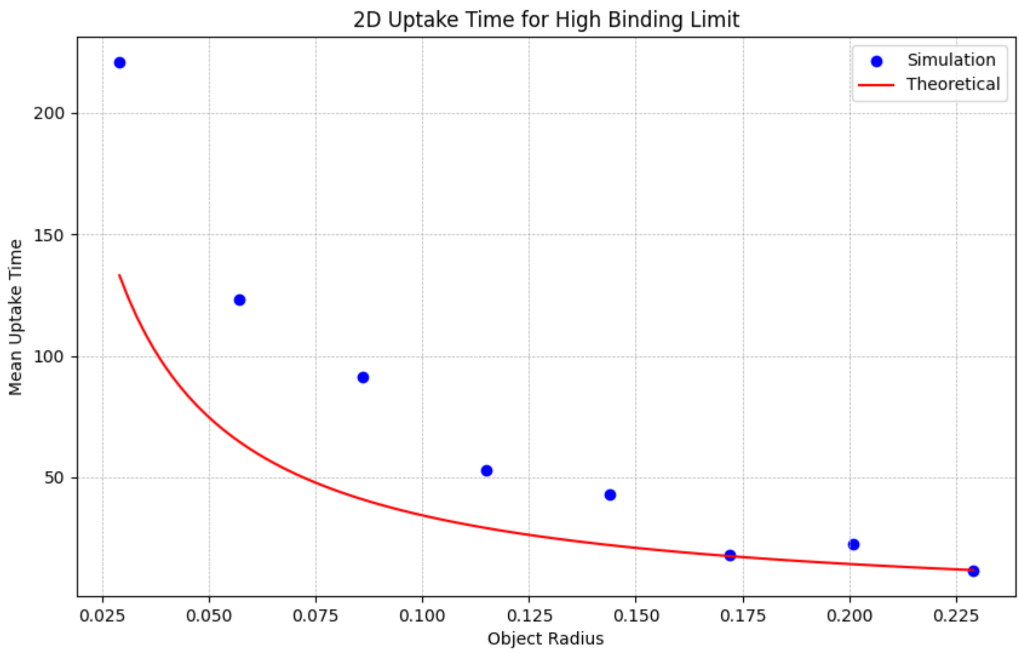

Figure 12: 2D problem with $k_{binding} \to \infty$ and $k_{unbinding} = 0$ limits. The theoretical curve is given by $T_{2D}(L) = (1/\alpha)(P_e(\xi L)v_e)^{-1}$ (see Sec.3.4.2). See Table3 in *Appendix* for the details of the simulation.

The systematic error that appears for the theoretical result might be explained as follows: The fraction $1/\alpha$ appears in the equation as a correction term to include the effect of angles which does not intersect with the object (Sec.3.4.2). This time scaling is valid only when the number of activation events is sufficiently high. This assumption appears to be too simple for the model. High radius values show less error since the effect from the angles of incidence is low.

5.2 Spherical Object Uptake in Finite Binding Rates

The ability of the pilus to uptake a spherical object was scrutinized under various binding and unbinding rates. The overall analysis is similar to the one in Sec.5.1.2 with the difference that k_{bind} and k_{unbind} have non-zero values. Additionally, in this section the effect of active binding region, L_{active}, (not just the tip of the pilus, but some finite length at the tip that activates k_{bind}, see Sec.4.3.3) was analyzed.

The mean uptake time for $L_{active} > 0$ case (see Sec.4.3.3) was analyzed. For different R_{object} and $k_{binding}$ values, the effect of active region was quantified by the simulations.

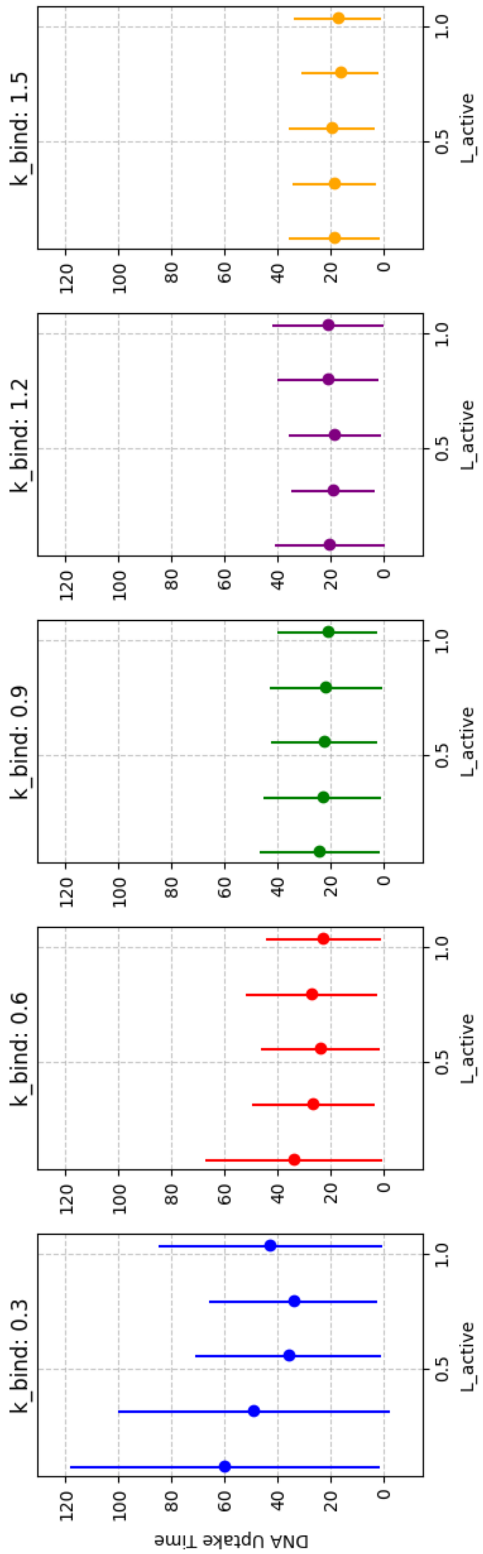

Figure 13: DNA uptake times in 2D setup for various different k_{bind} values. Increases in k_{bind} values decreases the effect of L_{active} on mean uptake time. See Table4 in *Appendix* for the details of the simulation.

It can be seen that an increase in L_{active} causes a decrease in uptake time. However, the effect of L_{active} on the mean uptake time decreases as k_{bind} increases. Such an effect of k_{bind} can be understood by realizing the way L_{active} decreases the mean uptake time:

The pilus tip may enter the region occupied by the object, continue extending, and even protrude from the other side of the object (see Fig.9). Since binding is a stochastic process in itself (Sec.4.3.2), when the pilus tip protrudes from the other side, it still has a chance of binding via L_{active}, as part of the active region of the pilus remains inside the object's region. The larger the L_{active} value, the higher the probability for pilus tip to bind even when the case of piercing through the object happens. However, when the k_{bind} value is large, the probability of piercing through decreases and pilus has much more tendecy to bind and carry the object with it. Therefore, the effect of large L_{active} on the mean uptake time decreases as k_{bind} increases.

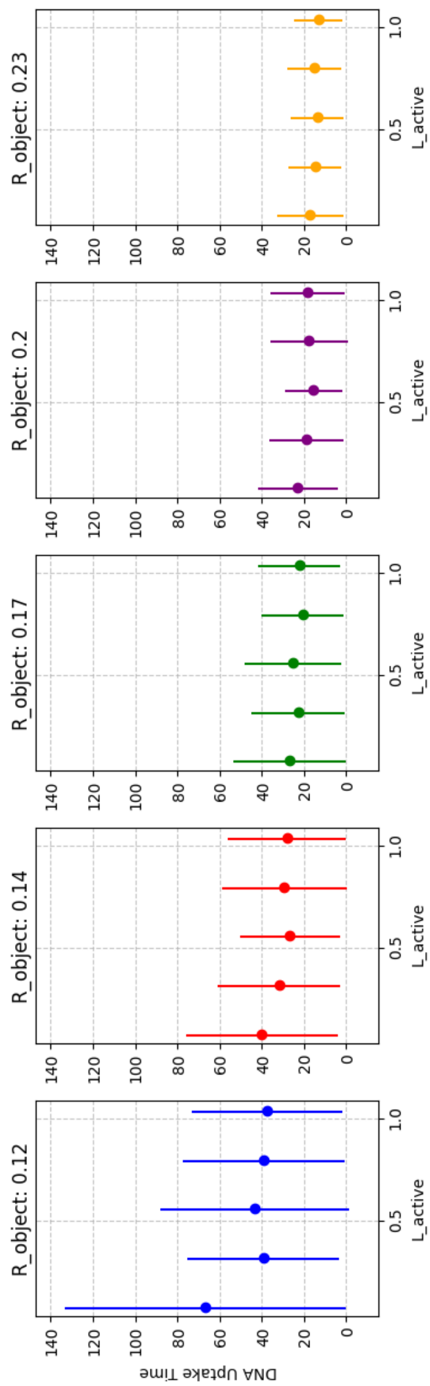

Figure 14: DNA uptake times in 2D setup for various different R_{object} values. Increases in R_{object} values decreases the effect of L_{active} on mean uptake time. See Table5 in *Appendix* for the details of the simulation.

Similar to the case of k_{bind} in Fig.13, R_{object} also decreases the effect of L_{active} on the mean uptake time. The reason is also similar. As R_{object} increases, the probability of piercing through the region decreases and before the piercing, there is enough time for pilus to switch to *bind* state. When, reached, the pilus will carry the object with it 4.3.2 and the size of L_{active} will not matter much.

5.3 Surface Adhesion

The ability of pili to bind to the surface and remain bound was investigated using the model given in Sec.4.4. The simulations were performed using the competitive binding model (Sec.4.2) for various F_{sens} and F_{stall} values. The time interval between the binding and unbinding of the pilus was considered as a single binding event, and the average binding time was used as a measure of binding ability.

5.3.1 Calculation of Mean Unbind Time

According to the model, only the retraction during binding affects the mean binding time. Suppose a binding occurs while the pilus is retracting. The equation for retract velocity for this model is (see Eq.63 and 64):

$$v_r(x) = v_{r0}\left(1 - \frac{\kappa x}{F_{stall}}\right) \tag{67}$$

where x is the amount of retraction in the absence of binding. Setting $\frac{dx}{dt} = v_r$ and solving gives:

$$x(t) = l_{max}(1 - e^{t/\tau}) \tag{68}$$

where $l_{max} := F_{stall}/\kappa$ and $\tau := l_{max}/v_{r0}$. The ficticious length does not exceed l_{max} and asymptotically approaches to it.

In order to have a quantitative understanding of the unbinding times, one requires the time evolution of the unbind rate. Substituting Eq.68 to 65 and defining $\xi := F_{sens}/\kappa$:

$$k_{unbind} = k^0_{unbind} \exp\left(\frac{\kappa l_{max}(1 - e^{t/\tau})}{\xi\kappa}\right) = k^0_{unbind} \exp(\alpha(1 - e^{t/\tau})) \tag{69}$$

where $\alpha := l_{max}/\xi = F_{stall}/F_{sens}$.

To determine the mean unbinding time, the probability distribution of unbinding needs to be found using time-dependent transition rates (see *Theory*):

$$p(t) = k_{unbind}(t) \exp\left(-\int_0^t k_{unbind}(t')dt'\right) \tag{70}$$

$$\langle T_{unbind,retract}\rangle = \int_0^\infty tp(t)dt = \int_0^\infty tk_{unbind}(t) \exp\left(-\int_0^t k_{unbind}(t')dt'\right)dt \tag{71}$$

It should be noted that this value is for the case where the pilus is retracting. For the model that was given, binding occurs even when the pilus is extendind or idle. In these situations, the expected value for mean unbinding time will be $\langle T_{unbind,nonretract}\rangle = 1/k^0_{unbind}$, since no effect on k_{unbind} will be present. Assuming that the expected time to change the length state is larger than $\langle T_{unbind}\rangle$, one can multiply the different unbinding times with the probability of being in *retract* state (φ) or not ($1-\varphi$). The probability φ can be found by:

$$\varphi = \frac{t_{retract}}{t_{retract} + t_{idle} + t_{extend}} \tag{72}$$

where t_i is the mean amount of time spent in state i. The corrected mean binding time will then be found as:

$$\langle T_{unbind,corrected}\rangle = \varphi\langle T_{unbind,retract}\rangle + (1-\varphi)\langle T_{unbind,nonretract}\rangle \tag{73}$$

It should not be forgotten that this is an approximation and not an unbiased value due to the assumption that for the cases where the length state changes happen during *bind* state, Eq.73 ignores the complicated change of $k_{unbind}(t)$ with time and acts linearly.

5.3.2 Simulation Results

The model that describes the surface binding simulated with competitive binding model (*Methods*) using different binding parameters. Fig.15 and Fig.16 belong to the same set of simulations and represent how F_{stall} and F_{bind} affect the mean binding time. The results match the calculations done in the previous section. The overall plateau of the mean binding time for increasing F_{stall} and F_{bind} can easily be predicted by Eq.69 (also the explanation).

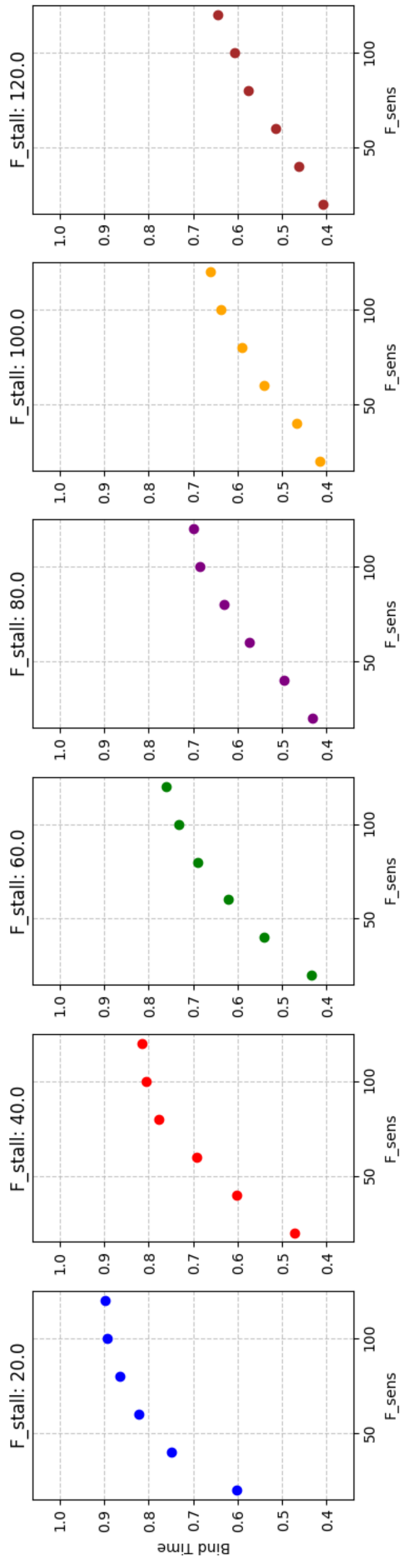

Figure 15: The mean binding time for pilus to the surface with respect to F_{sens} parameter. For high F_{stall} limits, k_{unbind} becomes less dependent on the tension force and approaches $1/k^0_{unbind}$ value. It can also be observed that for small F_{stall} values, the mean binding time reaches the plateau faster. Moreover, the value at which the mean binding time stabilizes decreases with increasing F_{stall}. See Table6 in *Appendix* for the details of the simulation.

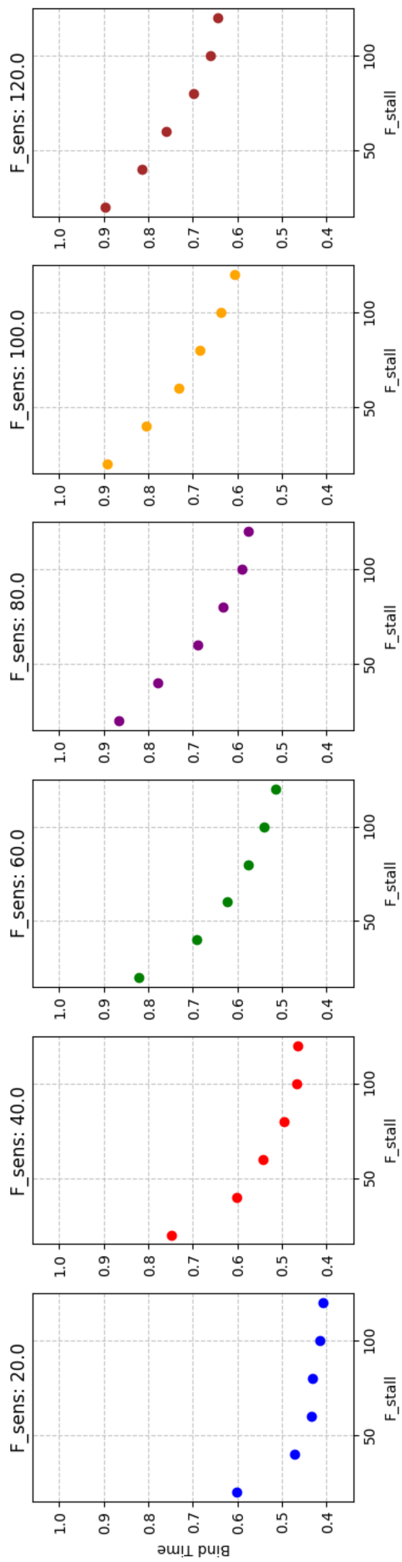

Figure 16: The mean binding time for pilus to the surface with respect to F_{stall} parameter. Large F_{stall} values causes the mean binding time to decrease and reach a plateau because the amount of maximum tension force is achieved at very low values and k_{unbind} does not change too much. See Table6 in *Appendix* for the details of the simulation.

It can be seen from Eq.71 that the main parameters affecting the mean unbinding time are $\alpha = F_{stall}/F_{sens}$ and $\tau = l_{max}/v_{r0}$. In order to see the quality of the predictions, the mean binding times for changing F_{sens} (not F_{stall} to ensure that τ remains the same) values for a constant $F_{stall} = 250$ (deliberately chosen to be high for clearer interpretation.) and were plotted by putting α values on the x axis. The correction mentioned for Eq.73 has to be applied to make sure the binding times coming from the *extension* and *idle* states are included.

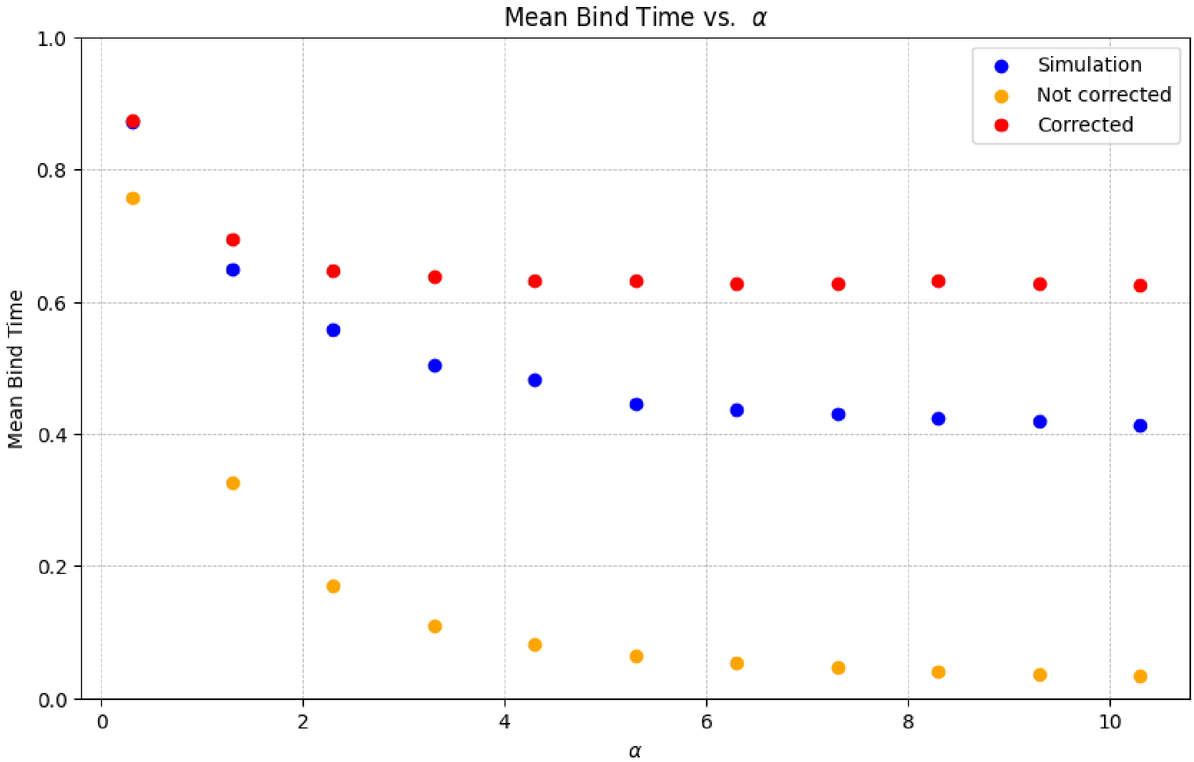

Figure 17: Comparison of theoretical (red for correction done at Eq.73 and orange for uncorrected mean binding time at Eq.71). The biased error is due to the fact that the correction ignores changes of length state during binding of the pilus tip by being a linear model even for short retraction times. (see text for explanation). This reasoning is supported by the observation that the correction does not cause an error for low α (high F_{stall}) regime where the binding time is small. See Table7 in *Appendix* for the details of the simulation.

The biased error that can be observed for high α (low F_{stall}) values can be explained by the correction method used at Eq.73. Such a model assumes that during the binding state, it does not change its length state, which is not so improbable considering the mean time spent in different length states and binding states.

To further explain the dependence of the mean binding time to α values, $\alpha = F_{stall}/F_{sens} = 1$ condition with various different values of $F_{stall}(F_{sens})$ were simulated.

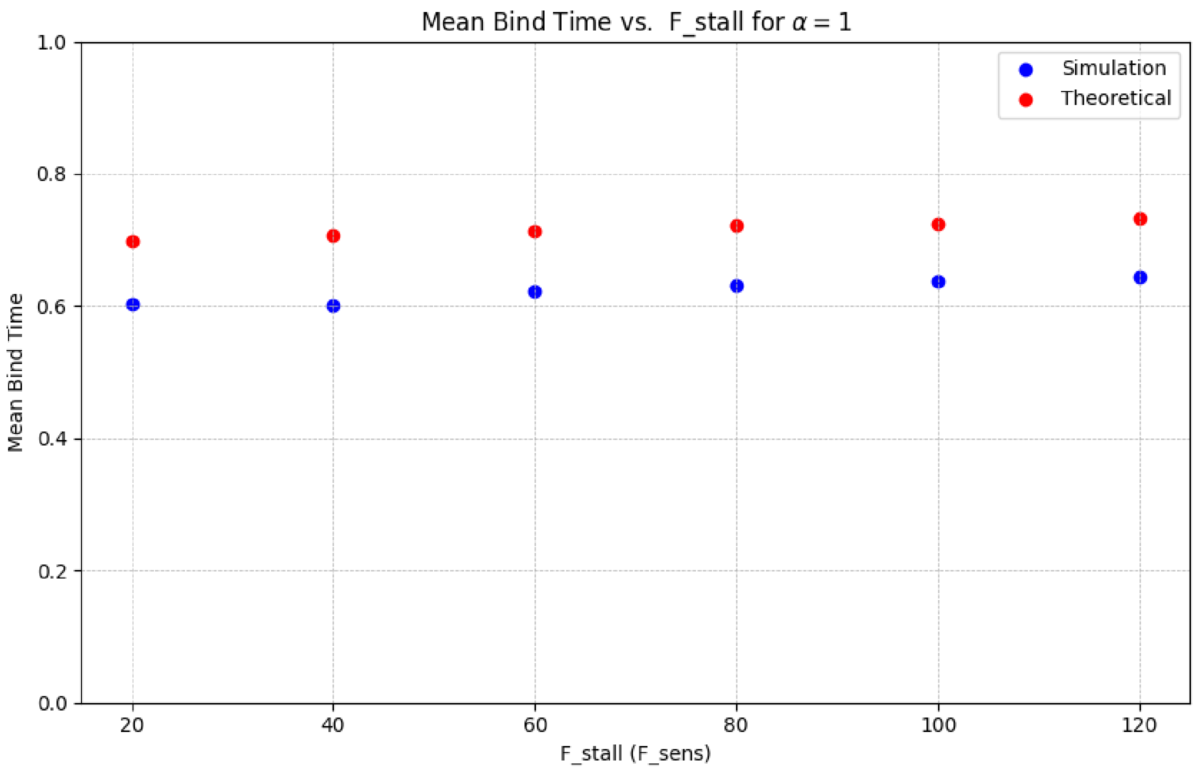

Figure 18: Mean binding times were simulated for constant $alpha = 1$ values for various different $F_{stall}(F_{sens})$ magnitudes. It can be seen that there is only a very small change on the mean binding time, indicating that mean binding time is primarily related to the relative proportions of the forces. Theoretical values are corrected with Eq.73. See Table8 in *Appendix* for the details of the simulation.

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The results show that although the τ values (which are proportional to F_{stall}) change, the overall effect on the mean binding time primarily comes from the α parameter.

5.4 Activation Time Distribution for Pilus

When pili become inactive (i.e., their length reaches zero), a certain amount of time is required for them to protrude again from the same site. In this chapter, the probability distribution for the pilus to be active again was analyzed for different models of re-activation.

In the paper by Koch et al. (2021), three main hypotheses were proposed regarding the binding of the PilT motor:

- Both binding and unbinding of PilT is independent of the pilus size (Stochastic Model)
- PilT unbinds stochastically and but does not bind again unless the pilus has nonzero length

- PilT unbinds immediately when the pilus length is zero and does not bind unless there is a pilus present

By employing the theoretical framework from chapter 3.2 with three states—retraction (PilT), idle, and extension (PilB)—and by making the extension state an absorbing state, one can determine the probability distribution for reaching the extension state for the first time.

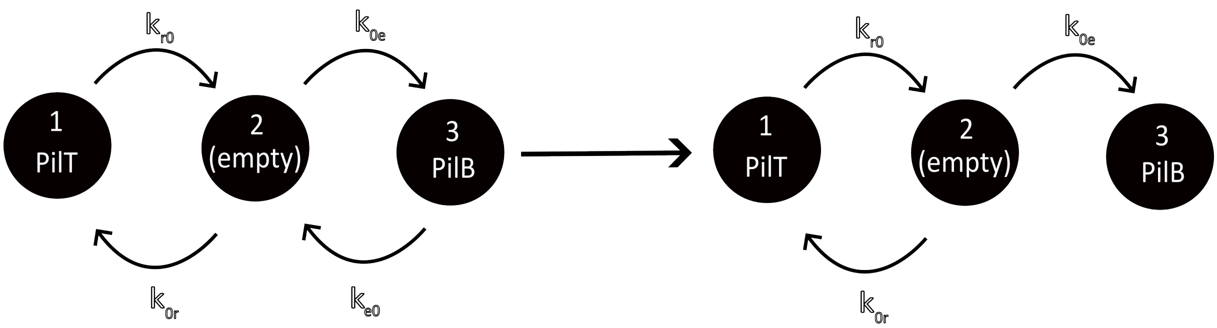

Figure 19: The 3-state modeling of the pilus motors. Transition rates out of the extension state was equated to zero to neglect the cases for the system to reach state 3 more than once (absorbing state). The states are enumerated for the interpretation with the matrix indicies.

Substituing the parameters from Simsek et al. (2023) and letting $k_{e0} = 0$, the transition matrix will be:

$$M_f = \begin{bmatrix} -k_{r0} & k_{0r} & 0 \\ kr_0 & -(k_{0r} + k_{0e}) & 0 \\ 0 & k_{0e} & 0 \end{bmatrix} = \begin{bmatrix} -0.11 & 2.5 & 0 \\ 0.11 & -2.9 & 0 \\ 0 & 0.4 & 0 \end{bmatrix} \tag{74}$$

The outcomes of these 3 different scenerios were analized:

5.4.1 Stochastic Model

In order to analyse the stochastic model, where the motor binding rates are independent of the pilus size, the initial probability distribution at Eq.19 must be $\vec{p_0} = (1, 0, 0)^T$, since in the moment which pilus length is zero, PilT must be the last motor that binds to the cite.
Taking the exponential of M_f from Eq.74 and substituing it to Eq. 13, one gets:

$$\vec{p}(t) = \begin{bmatrix} 0.032e^{-3.0t} + 0.97e^{-0.015t} \\ -0.037e^{-3.0t} + 0.037e^{-0.015t} \\ 1.0 + 0.0049e^{-3.0t} - 1.0e^{-0.015t} \end{bmatrix} \tag{75}$$

And applying Eq.18 will give:

$$S_1(t) = -0.0049e^{-3.0t} + 1.0e^{-0.015t} \tag{76}$$

where $S_3(t)$ is the probability for the system to not reach to state 3, namely, probability of non-activation in the time interval $(0, t)$.

5.4.2 PilT Does not Bind without Active Pilus

In this model, it was assumed that PilT does not immediately unbind when the pilus gets inactive but does not bind itself unless it gets active again. To achieve that, we further equate $k_{0r} = 0$ at Eq.74 for it to not bind again. The initial probability distribution will again be $\vec{p_0} = (1, 0, 0)^T$ as it were in the stochastic model.

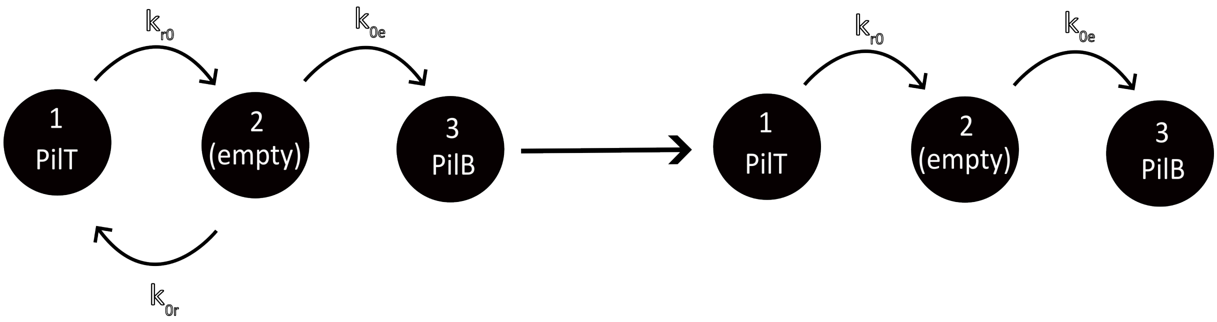

Figure 20: To satisfy the assumption that PilT does not bind to the pilus cite unless the pilus is active, $k_{0r} = 0$ was applied.

Putting everyting together in Eq.19:

$$S_2(t) = -0.38e^{-0.40t} + 1.4e^{-0.11t} \tag{77}$$

where $S_3(t)$ is the probability for the system to not reach to state 3, in the time interval $(0, t)$.

5.4.3 PilT Immediately Unbinds

In this model we not only assume that PilT does not bind to an inactive pilus but also immediately gets unbinds when the pilus gets inactive. The overall structure of the process is similar to the one in Fig.20 but this time, the initial probability distribution will be $\vec{p}(t) = (0, 1, 0)^T$, where initially the pilus is in the idle state.

Similar calculations will give:

$$S_3(t) = e^{-0.40t} \tag{78}$$

5.4.4 Comparison of the Models

In the paper from Simsek et al. (2023), pilus with zero length was taken as a different state, called the *inactive* state (see Sec.4), together with a rate from it to the *extending* state called k_{on}. The rate was found by experimental observations and it's value is provided to be $k_{on} = 0.36s^{-1}$ for the wild type Pseudomonas aeruginosa.

By including the three models from Koch et al. (2021) along with the *inactive* state approach using the rates provided in Simsek et al. (2023) (which were also used in this thesis), a comparison of the survival probabilities was conducted:

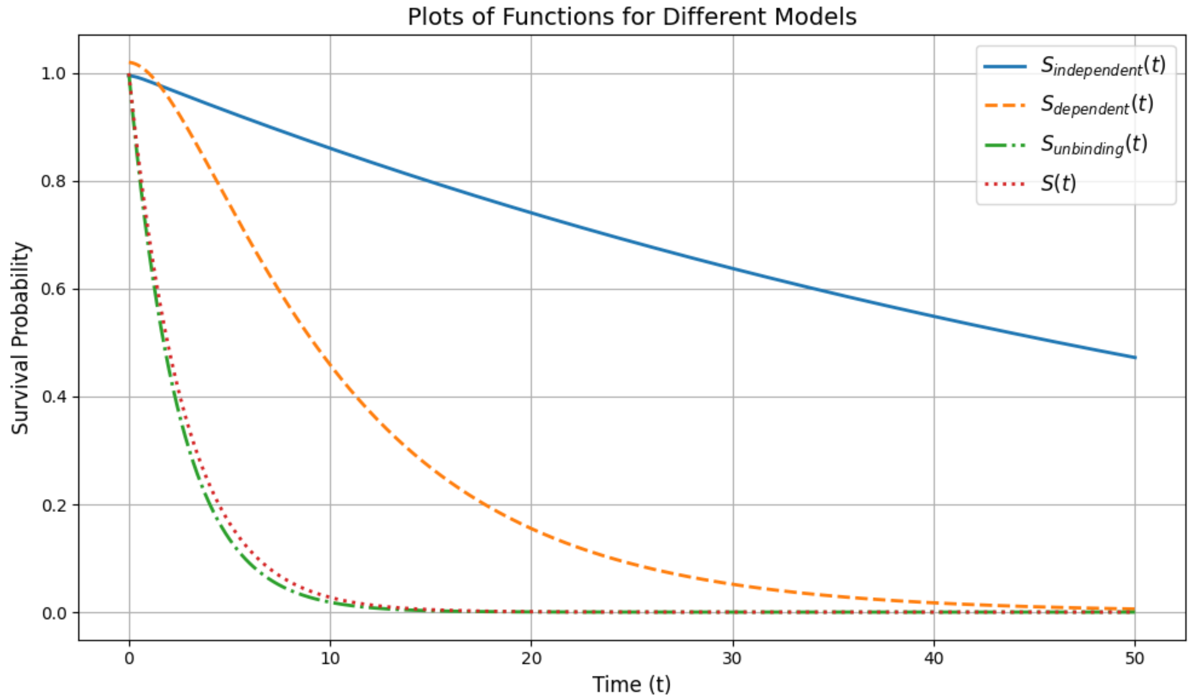

Figure 21: Comparison of different models. $S_{independent}$ for the model where binding of PilT is independent of the pilus (Eq.76), $S_{dependent}$ for the case where does not bind again (Eq.77), and $S_{unbinding}$ for the case where it immediately unbind and does not bind again (Eq.78). $S(t)$, on the other hand, is the model given by Simsek et al. (2023).

According to these results, the model presented in Simsek et al. (2023) aligns more closely with the $S_{unbinding}$ model, in which PilT immediately unbinds and does not bind again until the next activation (binding of PilB) occurs. This is contradicting with the results suggested by Koch et al. (2021), where it is argued that the data agrees better with the $S_{independent}$ model.

This disagreement may stem from differences in the observational methods used to measure activation times. The equilibrium length distribution of the pilus is known to decay exponentially (see 3.3.1). Therefore, many active pili have considerably small lengths. These lengths may be too small to be detected by the apparatus used in Koch et al. (2021), as mentioned in their paper.

6 Conclusions

T4P are extracellular appendages that play a crucial role in various cellular functions. In this thesis, their ability to uptake external molecules (DNA) and bind to surfaces is scrutinized. Additionally, the role of PilT in inactive pili was analyzed by comparing different models.

By modeling the pilus movement as a stochastic process in 1D and using Kramers' method, the mean time required for pilus tip to reach a certain length, L, was found to increase exponentially, $T(L) \propto e^{\kappa L}$. This result was then used to predict the uptake times for objects in 2D under various binding affinities, object sizes, and active binding region sizes. As expected, high binding affinities reduce the effect of active region lengths due to the immediate binding of the pili. A similar effect was observed for object radius: the larger the object, the greater the likelihood of pilus binding, regardless of its active region length. However, increasing the object size reduces the mean uptake time, as it increases the fraction of the incidence angle relative to the genesis angle. In other words, the pilus has a higher probability of selecting an angle that results in contact with the object.

For the surface adhesion analysis, the linear force-velocity relationship (Eq.64) was combined with the exponentially increasing force-rate model (Eq.65) to study mean binding times. It was shown that although the parameters F_{stall} and F_{sens} have an effect on the binding times as expected, the main parameter that affects the mean binding time is $\alpha = F_{stall}/F_{sens}$. For simulations where $F_{stall} = F_{sens}$, the mean binding time does not get affected much by the increasing $F_{stall}(F_{sens})$. This suggests that for bacterial species with high stall forces, the mean binding time (and thus the overall coordination of different pili) remains relatively unchanged on surfaces with high binding affinities.

Lastly, three different models for the behavior of PilT during the inactive state (Koch et al. (2021)) were examined by calculating the cumulative probability distributions for pilus to get active again. The results were compared to the model (and parameters) used in Simsek et al. (2023). In Koch et al. (2021), the pilus-independent model was supported by experimental observations. However, the survival probability distribution proposed by Simsek et al. (2023) aligned with the model in which PilT binding is strictly prohibited. Both studies focused on the same species, Pseudomonas aeruginosa. Both papers mentioned that the optical resolution of the microscope was insufficient to detect many short pili, making it difficult to accurately count active pili and determine the inactivation times.

A Appendix

A.1 Parameters

A.1.1 Table 1

Parameter	**Value**	**Unit**
Simulation		
Number of Simulations	1	
Total time	200.0	sec
Time step (dt)	0.0001	sec
Pilus		
Initial length	0.0	μm
Initial length state	inactive	
Velocities		
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
Length State Rates		
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}

Table 1: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.1.2 Table 2

Parameter	**Value**	**Unit**
Simulation		
Number of Simulations	200	
Total time	3000.0	sec
Time step (dt)	0.0001	sec
Pilus		
Initial length	0.0	μm
Genesis Angle	45.0	degrees
Initial length state	inactive	
Initial bind state	unbind	
Velocities		
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
Object		
Object initial distance (d)	[0.1, 0.6, 1.1, 1.6, 2.1, 2.6]	μm
Length State Rates		
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}
Bind Rates		
k_{bind}	∞	sec^{-1}
k_{unbind}	0.0	sec^{-1}

Table 2: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.1.3 Table 3

Parameter	**Value**	**Unit**
	Simulation	
Number of Simulations	400	
Total time	3000.0	sec
Time step (dt)	0.0001	sec
	Pilus	
Initial length	0.0	μm
Genesis Angle	45.0	degrees
Initial length state	inactive	
Initial bind state	unbind	
	Velocities	
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
	Object	
Object initial distance (d)	0.6	μm
Object radius (R)	[0.029, 0.057, 0.086, 0.115, 0.144, 0.172, 0.201, 0.229]	μm
	Length State Rates	
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}
	Bind Rates	
k_{bind}	∞	sec^{-1}
k_{unbind}	0.0	sec^{-1}

Table 3: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.1.4 Table 4

Parameter	**Value**	**Unit**
Simulation		
Number of Simulations	200	
Total time	3000.0	sec
Time step (dt)	0.0001	sec
Pilus		
Initial length	0.0	μm
Genesis Angle	45.0	degrees
Initial length state	inactive	
Initial bind state	unbind	
Active region length (L_{active})	[0.08,0.32,0.56,0.8,1.04]	μm
Velocities		
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
Object		
Object initial distance (d)	0.6	μm
Object radius (R)	0.2	μm
Length State Rates		
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}
Bind Rates		
k_{bind}	[0.3,0.6,0.9,1.2,1.5]	sec^{-1}
k_{unbind}	0.35	sec^{-1}

Table 4: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.1.5 Table 5

Parameter	**Value**	**Unit**
Simulation		
Number of Simulations	200	
Total time	3000.0	sec
Time step (dt)	0.0001	sec
Pilus		
Initial length	0.0	μm
Genesis Angle	45.0	degrees
Initial length state	inactive	
Initial bind state	unbind	
Active region length (L_{active})	[0.08,0.32,0.56,0.8,1.04]	μm
Velocities		
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
Object		
Object initial distance (d)	0.6	μm
Object radius (R)	[0.12,0.14,0.17,0.2,0.23]	μm
Length State Rates		
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}
Bind Rates		
k_{bind}	0.7	sec^{-1}
k_{unbind}	0.35	sec^{-1}

Table 5: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.1.6 Table 6

Parameter	**Value**	**Unit**
	Simulation	
Number of Simulations	1	
Total time	10000.0	sec
Time step (dt)	0.0001	sec
	Pilus	
Initial length	0.0	μm
Initial length state	inactive	
Initial bind state	unbind	
	Velocities	
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
	Length State Rates	
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}
	Binding Parameters	
k_{bind}	1.5	sec^{-1}
k_{unbind}	1.0	sec^{-1}
F_{stall}	[20.0, 40.0, $\cdots$, 120.0]	pN
F_{sens}	[20.0, 40.0, $\cdots$, 120.0]	pN

Table 6: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.1.7 Table 7

Parameter	**Value**	**Unit**
	Simulation	
Number of Simulations	1	
Total time	10000.0	sec
Time step (dt)	0.0001	sec
	Pilus	
Initial length	0.0	μm
Initial length state	inactive	
Initial bind state	unbind	
	Velocities	
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
	Length State Rates	
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}
	Binding Parameters	
k_{bind}	1.5	sec^{-1}
k_{unbind}	1.0	sec^{-1}
F_{stall}	250.0	pN
F_{sens}	$F_{stall}/[0.3, 1.3, \cdots, 10.3]$	pN

Table 7: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.1.8 Table 8

Parameter	**Value**	**Unit**
	Simulation	
Number of Simulations	1	
Total time	10000.0	sec
Time step (dt)	0.0001	sec
	Pilus	
Initial length	0.0	μm
Initial length state	inactive	
Initial bind state	unbind	
	Velocities	
Elongation speed (v_e)	0.42	μm/sec
Retraction speed (v_r)	0.55	μm/sec
	Length State Rates	
k_{on}	0.36	sec^{-1}
k_{0e}	0.6	sec^{-1}
k_{0r}	2.5	sec^{-1}
k_{e0}	0.95	sec^{-1}
k_{r0}	0.11	sec^{-1}
	Binding Parameters	
k_{bind}	1.5	sec^{-1}
k_{unbind}	1.0	sec^{-1}
F_{stall}	[20.0, 40.0, $\cdots$, 120.0]	pN
F_{sens}	F_{stall}	pN

Table 8: Simulation parameters were adopted from Simsek et al. (2023) and Koch et al. (2021).

A.2 Additional Plots

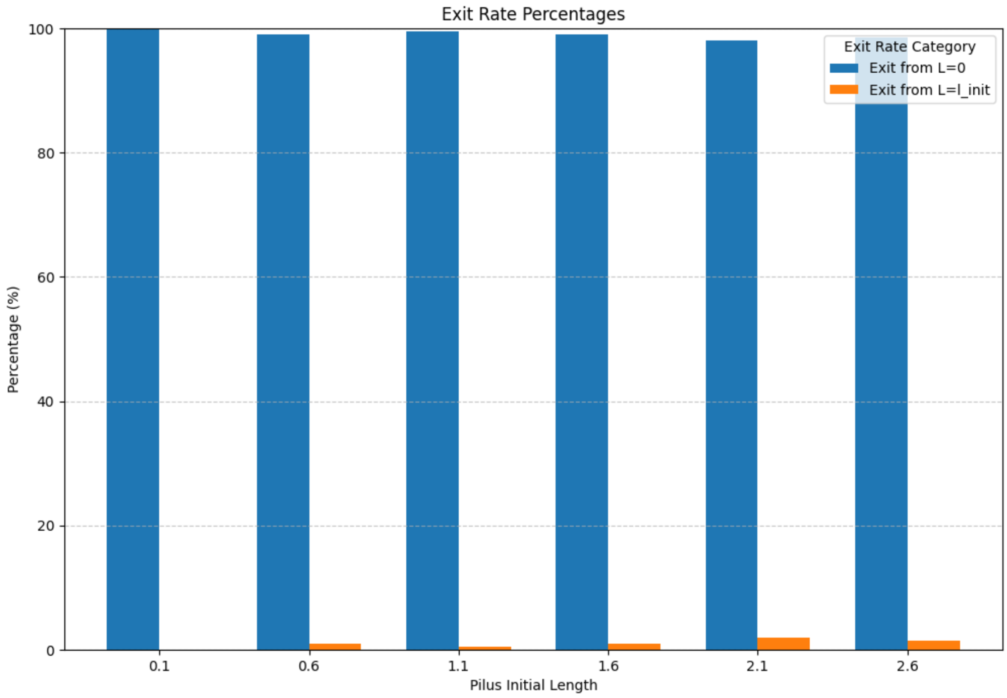

Figure 22: The probability of a pilus to start at a certain initial length at *retract* state and before going to pilus base, stop and extend back to the initial length. For length values around the average pilus length, only a small fraction returns back. This justifies the usage of Kramers' method without too much biased error.

B Electronic appendix

The source code used for the simulations can be accessed at GitHub repository:
https://github.com/kbozgul/MasterThesis

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Declaration of authorship

I hereby declare that this thesis is my own work, and that I have not used any sources and aids other than those stated in the thesis.

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